Trading with Richer Partners: Trade Integration and Structural Change in Mexico*†

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Abstract

Does the development characteristics of trading partners shape how trade integration drives structural change in a developing economy? Using a multi-country, three-sector Ricardian model calibrated to Mexico, the United States, Canada, and the rest of the world over 1971–2018, this paper examines how partner development characteristics condition the effects of trade integration. By 2018, Mexico's observed trade-cost reductions increase its manufacturing value-added share by 8.49 percentage points relative to constant trade-costs predictions, compared with only 1.88 points when its North American partners are assigned Mexican development characteristics. Conversely, the agricultural value-added effect increases from 0.02 to 8.02 points. Manufacturing employment rises by 6.71 points with actual partners but falls by 1.35 points with similarly developed partners. A decomposition of partner characteristics identifies technology differences as the dominant contributor, with relative sectoral productivity profiles playing a substantially larger role than overall productivity levels. These findings establish that the structural consequences of trade integration depend critically on trading partners' sectoral productivity profile, rather than their aggregate development levels alone.

KEYWORDS: Structural Change; International Trade; Trade Integration; Economic Development; Trade Liberalization.
JEL CODES: F14; F15; O14; O19; O24.

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1 Introduction

Trade integration changes the allocation of economic activity across agriculture, manufacturing, and services. By expanding access to foreign markets and imported inputs, it can support manufacturing production and facilitate movement out of agriculture. At the same time, stronger import competition and changes in relative prices can reinforce specialization in other sectors. The direction of structural change therefore depends on the productive capabilities and demand patterns of the economies connected by integration.[1] This paper asks how the sectoral response to a given trade-cost path changes when a developing economy's trading partners have different development characteristics.

This question is particularly relevant in an increasingly integrated global economy. Over the past four decades, international trade integration has expanded substantially through regional and multilateral trade agreements. According to the World Trade Organization (WTO), 388 notified regional trade agreements were in force as of September 2026, many connecting developing economies with substantially more advanced partners.[2] Integration with a richer economy provides access to a larger foreign market but also exposes domestic producers to competition from firms operating under different productivity and cost conditions. Differences in sectoral productivity and expenditure patterns shape comparative advantage and the composition of foreign demand, thereby influencing the allocation of economic activity across sectors. Consequently, the role of trade integration on structural change cannot be inferred from changes in trade barriers alone. They also depend on the development characteristics of the economies being integrated. Understanding this interaction is essential to explaining why comparable reductions in trade barriers can generate different patterns of structural change.

This paper examines how the development characteristics of trading partners shapes the effects of trade integration on structural change, focusing on Mexico's integration with the United States and Canada. Mexico provides a particularly informative setting for this analysis. Since the 1970s, its economy has undergone substantial structural change while transitioning from relative trade isolation to deep integration with global markets. Two major policy milestones marked this transition: Mexico's accession to the General Agreement on Tariffs and Trade (GATT) in 1986 and the implementation of the North American Free Trade Agreement (NAFTA) in 1994. Importantly, NAFTA deepened Mexico's integration with two developed economies, the United States and Canada, with the former remaining Mexico's dominant trading partner. As documented in Section 2, Mexico's trade expansion was concentrated in North America, particularly in exports to the United States, rather than distributed evenly across trading partners. This integration exposed Mexico to the foreign demand, production costs, and sectoral productivity patterns of substantially more advanced economies. The Mexican experience therefore provides an opportunity to quantify how integration with richer trading partners shapes sectoral specialization and labor reallocation, and how these effects would differ if integration occurred with economies with similar development characteristics.

To conduct this analysis, I use an established multi-country, three-sector Ricardian general equilibrium model of structural change. The model features agriculture, manufacturing, and services, and incorporates non-homothetic preferences, such that the income elasticities of demand for sectoral goods differ from one (e.g., Kongsamut et al. 2001, Herrendorf et al. 2013, Uy et al. 2013). Sector-specific productivity growth generates endogenous changes in relative prices, thereby inducing structural change (e.g., Baumol 1967; Ngai & Pissarides 2007). The model further introduces international trade in intermediate inputs, which affects the sectoral composition of production and employment. Varieties in all three sectors can be sourced inter-

[1] Structural change is defined as the reallocation of economic activity across the three broad sectors of the economy, agriculture, manufacturing, and services, that accompanies the process of modern economic growth (Kuznets 1973, Herrendorf et al. 2014).

[2] World Trade Organization, "Regional Trade Agreements Database," RTA Tracker and summary tables, accessed August 2026: https://rtais.wto.org/UI/PublicMaintainRTAHome.aspx.

nationally, while final sectoral composite goods are assembled domestically from those traded varieties. The patterns of specialization and international trade are determined by relative productivity differences across countries, as in Uy et al. (2013), Sposi (2019) and Sposi et al. (2026).

Trade integration affects structural change through three interrelated channels. First, reductions in bilateral trade costs alter relative competitiveness and trade shares, encouraging countries to specialize according to their comparative advantage. Changes in domestic sourcing and foreign demand consequently affect sectoral employment and value-added shares. Second, trade integration changes real income, generating shifts in sectoral expenditure through nonhomothetic preferences. As income rises, expenditure shifts toward sectors with higher income elasticities. Third, lower trade costs, particularly for intermediate inputs, alter sectoral production costs and relative prices. These price changes affect expenditure patterns and induce further reallocation across sectors. Together, specialization, income effects, and relative-price adjustments determine how trade integration shapes structural change.

The magnitude and direction of these effects depend on the development characteristics and market size of trading partners. Differences in sectoral productivity, wages, and intermediate-input costs shape comparative advantage and thus the sectoral response of trade flows to lower trade costs. In particular, a larger productivity gap between Mexico and an advanced partner in agriculture than in manufacturing can strengthen Mexico's comparative advantage in manufacturing relative to agriculture, encouraging specialization in manufacturing as integration deepens. On the demand side, nonhomothetic preferences imply that differences in income across trading partners generate differences in sectoral expenditure patterns, while partner market size determines the scale of potential foreign demand. These supply- and demand-side forces interact with changes in relative prices and domestic expenditure, potentially reinforcing or offsetting one another. Consequently, identical reductions in trade costs can generate different patterns of sectoral reallocation depending on the development characteristics of the economies being integrated.

I calibrate the model to key features of the Mexican, Canadian, and U.S. economies over 1971–2018, using data on sectoral employment, national accounts, trade flows, and prices. I then conduct counterfactual experiments to quantify how trading partners' development characteristics shape the effects of trade integration on structural change. The identification strategy compares the effects of the same trade-cost changes across alternative partner-development environments. Within each environment, I measure the effects of trade integration by comparing sectoral allocations under observed trade costs with those under counterfactual paths that hold Mexico's bilateral trade costs fixed at their initial, pre-GATT, or pre-NAFTA levels. Comparing these effects across environments isolates the role of partner characteristics in the response to trade integration, rather than differences in sectoral allocations that would arise even without subsequent changes in trade costs.

The analysis proceeds in three steps. First, I construct an alternative environment in which the United States and Canada are assigned Mexico's initial sectoral technology levels, sectoral wage wedges, and preference weights, while preserving each partner's subsequent sector-specific technology growth. Second, I vary these groups of characteristics separately and jointly and apply an order-independent decomposition to quantify their respective contributions to differences in the effects of trade integration. Third, I distinguish changes in overall partner productivity from changes in its relative sectoral composition and trace the response across intermediate partner configurations.

The main quantitative finding is that partner development shapes both the magnitude and the sectoral direction of the effects of trade integration. Trading with advanced partners yields a smaller agricultural share and a larger manufacturing share than an otherwise comparable environment with Mexico-level partners, with the U.S. relationship accounting for the larger separate partner effect. By 2018, trade integration increases Mexico's manufacturing value-

added share by 8.49 percentage points and reduces the services share by 8.51 points relative to constant trade costs counterfactual. With Mexico-level partners, the same trade-cost path raises manufacturing value added by 1.88 points and lowers services by 9.90 points, while agriculture increases by 8.02 points rather than 0.02 points. The agricultural effect is positive in both environments. By contrast, the manufacturing employment effect changes from an increase of 6.71 points to a decline of 1.35 points. Thus, partner development substantially alters the sectoral incidence of trade integration, with different implications for employment and nominal value added.

The GATT and NAFTA trade-cost-window experiments further show that identical changes in trade cost path not generate the same sectoral allocation across partner environments. The post-1993 North American trade-cost path raises Mexico's manufacturing employment share by 2.12 percentage points with actual partners and 0.70 points with Mexico-level partners; the corresponding value-added effects are 2.13 and 0.75 points. Both manufacturing responses remain positive in this window.

The development-characteristics bundle captures several dimensions along which trading partners differ in their levels of development, including sectoral technology, wage wedges, and preference weights. To assess their relative importance, I decompose the contribution of partner development into these components. The results show that differences in technology account for most of the variation in Mexico's structural response to trade integration across partner configurations. I further distinguish between two dimensions of technology: the overall level of productivity, which influences production costs, incomes, and purchasing power, and the relative sectoral productivity profile, which shapes comparative advantage and specialization. The counterfactual results indicate that the latter is the dominant mechanism. Thus, the effects of trade integration depend not simply on trading partners' aggregate productivity or income levels, but critically on the distribution of their productivity advantages across sectors.

Taken together, these findings establish that trading partners' development characteristics are not merely scale factors but fundamental determinants of how trade integration affects structural change. Differences in sectoral productivity, wage wedges, and expenditure patterns shape the responses of foreign demand, production costs, and relative prices to reductions in trade barriers, thereby influencing the direction and magnitude of sectoral reallocation. In particular, the results highlight the central role of partners' sectoral productivity profile in determining whether trade integration promotes manufacturing expansion or reallocates economic activity toward agriculture.

Related Literature. The paper contributes to the literature on trade and structural change. Matsuyama (2009), Uy et al. (2013), Swiecki (2017), Betts et al. (2017), Teignier (2018), Sposi (2019), and Cravino & Sotelo (2019) emphasize mechanisms through which trade affects sectoral allocations. More recently, Sposi et al. (2026) study the interaction of trade and sector-biased technological change in premature deindustrialization and industry polarization. De Vries et al. (2026) examine structural change in sub-Saharan Africa and developing Asia, emphasizing domestic and foreign sectoral productivity and the transmission of foreign productivity through trade. The present paper complements this work by comparing responses to the same trade-cost path across partner configurations and decomposing the underlying technology, wage-wedge, and preference structures.

This paper highlights an additional determinant of structural change: the development characteristics of trading partners, particularly their relative sectoral productivity profiles and sectoral wage wedges. By identifying how these characteristics shape the effects of trade integration on sectoral reallocation, the paper extends the literature on the drivers of structural change beyond the conventional income, relative-price, and trade channels (Ilyina & Samaniego 2012, Liu 2017, Van Neuss 2019, Skott 2021, Grabowski & Self 2024, Chakrabarti et al. 2025).

This paper is also related to the quantitative and dynamic literature on structural change,

which emphasizes the roles of sectoral productivity, preferences, intermediate input, and factor-market mechanisms. Contributions to this literature include Foellmi & Zweimüller (2008), Buera & Kaboski (2009), Alvarez-Cuadrado & Poschke (2011), Guilló et al. (2011), Dekle & Vandenbroucke (2012), Herrendorf et al. (2015), Padilla-Pérez & Villarreal (2017), Chen (2020), Lewis et al. (2022) and Avoumatsodo & Leunga Noukwe (2025).

The interpretation of partner characteristics also relates to the literature on income and bilateral trade. Fieler (2011) shows how nonhomothetic demand and production characteristics help explain trade patterns, while Caron et al. (2014) connect sectoral income elasticities with skill intensities. In these settings, richer trading partners differ in both demand and supply. My experiments distinguish income-induced demand changes under fixed preferences from changes in country-specific preference weights, and distinguish the level of partner technology from its sectoral composition. This distinction is also relevant to Fujiwara & Matsuyama (2024), who show how sector-specific technology adoption lags can generate different industrialization paths.

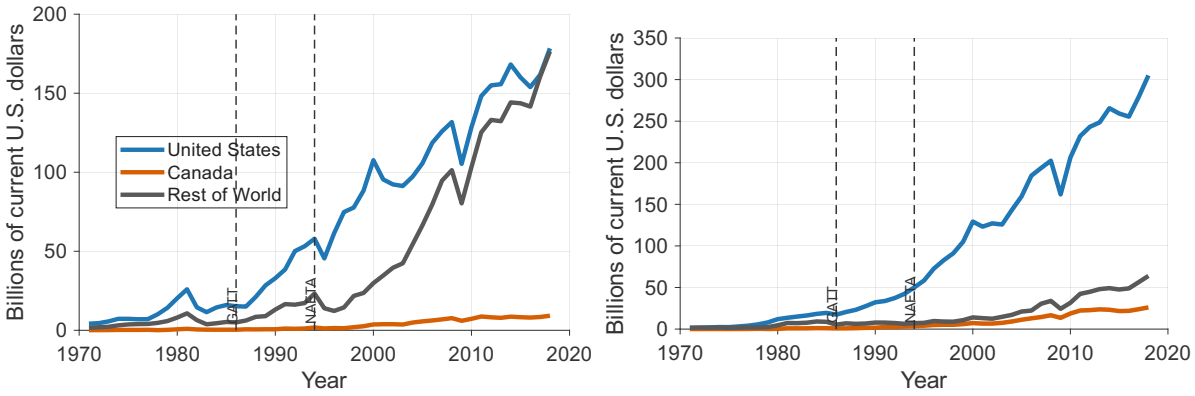
A third connection concerns agricultural specialization and industrialization. Matsuyama (1992) demonstrates why the relationship between agricultural productivity and development depends on openness and comparative advantage. Tombe (2015) studies agricultural trade barriers and the allocation of labor, while Teignier (2018) emphasizes agricultural imports in structural transformation. These contributions provide a basis for examining how partner agricultural productivity affects the home country's response to integration . They also make clear that movement out of agriculture and expansion of manufacturing are distinct outcomes. The results are interpreted in relation to the evidence on premature deindustrialization in Rodrik (2016) and the cross-country evidence on agricultural development in Daymard (2025).

Finally, this paper is related to the literature examining the effects of trade agreements, particularly NAFTA, on the Mexican economy. Hanson (1997) documents the importance of market access for the regional distribution of Mexican manufacturing employment and wages. Park (2001) finds that NAFTA increased bilateral trade between Mexico and the United States in vehicles, machinery, and iron and steel products. McDaniel & Agama (2003) shows that U.S. import demand for Mexican goods responded strongly to reductions in U.S. tariffs on Mexican products between 1989 and 2001, with larger responses in the post-NAFTA period. By contrast, Konno & Fukushige (2003) finds no additional long-run effect of NAFTA on bilateral trade between Mexico and the United States, while Caliendo & Parro (2015) quantify NAFTA's trade and welfare effects in a Ricardian model with intermediate inputs and sectoral linkages. The present analysis focuses on national employment and nominal value-added shares and on their sensitivity to partner characteristics. More importantly, I distinguish the effects of trade integration from the role played by differences in development levels across trading partners, showing how partner development characteristics shapes the sectoral reallocation generated by trade integration.

The remainder of the paper proceeds as follows. Section 2 documents the motivating trade facts. Section 3 introduces the model and provides theoretical insights into the mechanisms linking trade integration and structural change. Section 4 describes the data, calibration strategy, and model fit. Section 5 presents the counterfactual results. Section 7 concludes.

2 Motivating Facts: Mexico's Trade Concentration

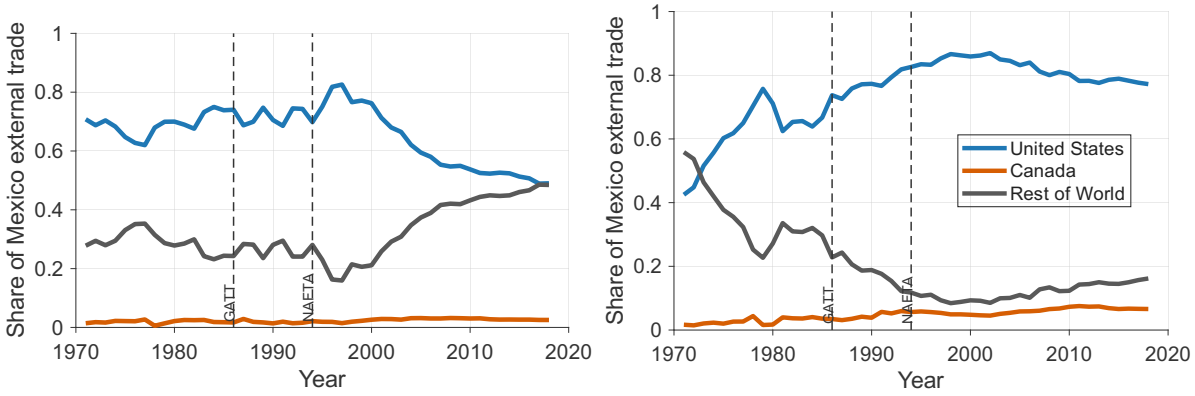
Two facts about Mexico's external trade motivate the quantitative emphasis on partner development. First, Mexico's integration was not only an increase in overall openness; it was also a reorientation toward a specific set of advanced partners. Figure 1 plots Mexico's total imports and exports with the United States, Canada, and the rest of the world, with vertical lines marking GATT accession in 1986 and NAFTA in 1994. Trade with the United States expands sharply over the sample, especially on the export side. Mexican exports to the United States rise from 1.26 billion current U.S. dollars in 1971 to 17.35 billion in 1986, 50.11 billion



(a) Import (b) Export

Figure 1: Mexico's Bilateral Trade with Main Partners

Notes: The figure reports Mexico's total imports from, and exports to, the United States, Canada, and the rest of the world. Values are in billions of current U.S. dollars. Vertical lines mark Mexico's GATT accession in 1986 and NAFTA in 1994.



(a) Import shares (b) Export shares

Figure 2: Partner Shares in Mexico's External Trade

Notes: The figure reports the shares of the United States, Canada, and the rest of the world in Mexico's external imports and exports. Domestic absorption is excluded from the denominator. Vertical lines mark Mexico's GATT accession in 1986 and NAFTA in 1994.

in 1994, and 305.01 billion in 2018. Imports from the United States also expand substantially, although by 2018 imports from the rest of the world are nearly as large as imports from the United States.

Second, the composition of Mexico's trading partners is highly asymmetric. Figure 2 shows that the United States accounted for 42.5 percent of Mexico's exports in 1971, 73.7 percent in 1986, 82.6 percent in 1994, and 77.2 percent in 2018. Canada remained a smaller trading partner, although its export share increased from 1.7 percent in 1971 to 6.6 percent in 2018. On the import side, the United States remained dominant throughout the NAFTA period, while by 2018 the United States and the rest of the world each accounted for roughly one-half of Mexico's external imports. These patterns are central to the question studied in this paper. If Mexico's structural change depends on foreign demand, the cost of imported inputs, and the productivity of its trading partners, then the substantial development gap between Mexico and its two main North American partners may be an important determinant of how trade integration affects sectoral reallocation. This paper therefore asks how Mexico's structural change would have differed had it integrated with trading partners at a similar level of development rather than with the more advanced United States and Canada.

These facts establish Mexico's exposure to a particular set of trading partners. They do not, by themselves, identify how partner characteristics change the sectoral response to trade integration. The quantitative experiments below address this conditional question within a calibrated general equilibrium framework.

3 Model

I use a multi-country, multi-sector Ricardian general equilibrium framework to examine how the sectoral effects of trade integration depend on the characteristics of trading partners. The model builds on Eaton & Kortum (2002), Caliendo & Parro (2015), and Sposi et al. (2026). There are four economies, indexed by $i, j \in \mathbb{I} = \{\text{MEX}, \text{CAN}, \text{USA}, \text{ROW}\}$, and three sectors, indexed by $k, n \in \{a, m, s\}$. Each country is populated by a representative household with nonhomothetic preferences and firms operating under constant returns to scale. The model integrates non-unitary income and substitution elasticities of demand alongside sector-specific productivity growth. This formulation allows both the Engel effect and the Baumol effect to operate simultaneously within the economy. Each sector produces a continuum of varieties using labor and intermediate inputs. In the benchmark specification, all varieties are tradable across countries, and international trade is subject to iceberg trade costs.

3.1 Production and International Sourcing

Sector k in country i produces a continuum of varieties $z \in [0, 1]$ using labor and intermediate inputs. The production technology is

$$Y_{ik}(z) = a_{ik}(z) L_{ik}(z)^{1-\lambda_{ik}} \left[\prod_{n \in \{a,m,s\}} M_{ikn}(z)^{\gamma_{ikn}} \right]^{\lambda_{ik}}, \qquad \sum_{n \in \{a,m,s\}} \gamma_{ikn} = 1, \tag{1}$$

where λ_{ik} is the intermediate-input share and γ_{ikn} is the share of sector n in the intermediate bundle used by sector k, $Y_{ik}(z)$ denotes output, $L_{ik}(z)$ is labor input, and $M_{ikn}(z)$ is the quantity of composite good from sector n used as an intermediate input in the production of the sector k good. Productivity, a_{ik}, draws follow the Fréchet distribution $F_{ik}(a) = \exp(-T_{ik} a^{-\theta})$. The parameter T_{ik} is a technology scale; $T_{ik}^{1/\theta}$ is proportional to a common scale of the productivity draws.

Let P_{in} denote the price of the sector-n composite good in country i. Under perfect competition, the unit cost excluding the variety-specific productivity draw is

$$v_{ik} = \Upsilon_{ik} w_{ik}^{1-\lambda_{ik}} \prod_{n \in \{a,m,s\}} P_{in}^{\lambda_{ik}\gamma_{ikn}}, \qquad \Upsilon_{ik} = (1-\lambda_{ik})^{\lambda_{ik}-1} \lambda_{ik}^{-\lambda_{ik}} \prod_{n \in \{a,m,s\}} \gamma_{ikn}^{-\lambda_{ik}\gamma_{ikn}}. \tag{2}$$

Sectoral wages satisfy $w_{ik} = \xi_{ik} w_{im}$, with $\xi_{im} = 1$ as in Buera & Kaboski (2009) and Swiecki (2017). These wedges capture labor-market frictions and distortions permit employment and value-added shares to differ. They are reduced-form accounting objects rather than separate estimates of particular labor-market distortions.

Composite goods aggregate varieties with a CES elasticity $\eta > 1$ and can be used for final consumption or as intermediate inputs. All sectors are internationally tradable. Country i sources each variety from the supplier with the lowest delivered cost, $\tau_{ijk} v_{jk} / a_{jk}(z)$, where the first country index denotes the destination. Domestic trade costs are normalized to one. The resulting price index is given ass follows

$$P_{ik} = G \left[\sum_{j=1}^{4} T_{jk} (v_{jk} \tau_{ijk})^{-\theta} \right]^{-1/\theta}, \tag{3}$$

$$\tag{4}$$

where $G = \Gamma(1 + (1-\eta)/\theta)^{1/(1-\eta)}$ and $\eta < 1 + \theta$. The share of expenditure that country j allocates to imports of sector-k goods from country i, denoted π_{jik}, is given by:

$$\pi_{jik} = \frac{T_{ik} \left(v_{ik}\tau_{jik}\right)^{-\theta}}{\sum_{\ell=1}^{4} T_{\ell k} \left(v_{\ell k}\tau_{j\ell k}\right)^{-\theta}}. \tag{5}$$

To aggregate variety-level production and inputs to the sectoral level, I integrate across all varieties:

$$L_{ik} = \int_0^1 L_{ik}(z)\, dz, \quad M_{ikn} = \int_0^1 M_{ikn}(z)\, dz, \quad Y_{ik} = \int_0^1 Y_{ik}(z)\, dz. \tag{6}$$

Here, L_{ik} denotes total labor used in sector k, M_{ikn} is the total quantity of inputs from sector n used in sector k, and Y_{ik} is the total output of sector k.

3.2 Households

Preferences follow the nonhomothetic CES specification of Comin et al. (2021). The consumption aggregator is implicitly defined by :

$$\sum_{k\in\{a,m,s\}} \omega_{ik} \left(\frac{C_i}{L_i}\right)^{\frac{1-\sigma}{\sigma}\varepsilon_k} \left(\frac{C_{ik}}{L_i}\right)^{\frac{\sigma-1}{\sigma}} = 1 \tag{7}$$

where C_i is a composite index of sectoral consumption in agriculture, manufacturing, and services in country i and L_i is the labor endowment. The parameter $\sigma > 0$ governs substitution across sectoral goods, and $\varepsilon_k > 0$ governs the nonhomothetic Engel curve while parameters ω_{ik} are sectoral preference weights.

Following Caliendo et al. (2018) and Sposi et al. (2026), I abstract from international borrowing and lending by modeling trade imbalances as international transfers. Specifically, an exogenous fraction $\phi_i \in [0, 1]$ of country i 's income is contributed to a global portfolio. This lobal pool is then redistributed uniformly across countries in the form of a per-capita lump-sum transfer $\mathcal{T}$, ensuring that world trade remains balanced in each period. Under this assumption, Household expenditure satisfies

$$E_i = (1-\phi_i)Y_i + L_i\mathcal{T}, \qquad \mathcal{T} = \frac{\sum_j \phi_j Y_j}{\sum_j L_j}. \tag{8}$$

where Y_i is the national income, given by $Y_i = \sum_k w_{ik}L_{ik}$. World transfers sum to zero, and net exports are

$$NX_i = \phi_i Y_i - L_i\mathcal{T}, \tag{9}$$

It is worth nothing that holding ϕ_i fixed in a counterfactual does not fix the level of net exports, since incomes are endogenous.

Solving the minimization problem of household yields the following demand function for sector-k consumption:

$$\frac{C_{ik}}{L_i} = \omega_{ik}^{\sigma} \left(\frac{P_{ik}}{P_i}\right)^{-\sigma} \left(\frac{C_i}{L_i}\right)^{\varepsilon_k(1-\sigma)+\sigma}, \tag{10}$$

where the consumption-based price index P_i is given by:

$$P_i = \left[\sum_{k\in\{a,m,s\}} \omega_{ik}^{\sigma} P_{ik}^{1-\sigma} \left(\frac{C_i}{L_i}\right)^{(1-\sigma)(\varepsilon_k-1)}\right]^{\frac{1}{1-\sigma}}. \tag{11}$$

3.3 Equilibrium

The model economy is characterized by a set of time-invariant parameters $(\varepsilon_k, \sigma, \theta, \eta, \lambda_{ik}, \gamma_{ijk}, \omega_{ik})$ and a collection of time-varying exogenous processes, including sectoral technology scales, bilateral trade costs and sectoral wage wedges $\{T_{ik}, \tau_{ijk}, \xi_{ik}\}$, labor endowments $\{L_i\}$, and trade imbalance parameters $\{\phi_i\}$.

Definition. A competitive equilibrium in this framework is a sequence of sectoral goods, factor prices and aggregate price $\{P_{ia}, P_{im}, P_{is}, w_{ia}, w_{im}, w_{is}, P_i\}_{i\in\mathbb{I}}$, where $\mathbb{I} = \{1, 2, 3, 4\}$, allocations $\{L_{ia}, L_{im}, L_{is}, Q_{ia}, Q_{im}, Q_{is}, C_{ia}, C_{im}, C_{is}, C_i, \mathcal{T}\}_{i\in\mathbb{I}}$, and sectoral trade shares $\{\pi_{ija}, \pi_{ijm}, \pi_{ijs}\}_{i,j\in\mathbb{I}}$, such that, given prices and the allocations (i) Firms maximize profits, taking prices as given, according to the production technology specified in Equation (1); (ii) Households choose sectoral consumption to maximize utility subject to the budget constraint (8) and preference structure defined by Equations (7); (iii) Each country sources every variety from the lowest-cost supplier, as in Equation (5); and (iv) All factor and goods markets clear.

Market Clearing Conditions. The factor market clearing conditions for labor and sectoral composite goods in each period are as follows:

$$L_i = \sum_{k\in\{a,m,s\}} L_{ik} \tag{12}$$

$$Q_{ik} = C_{ik} + \sum_{n\in\{a,m,s\}} M_{ink}. \tag{13}$$

Let $X_{ik} = P_{ik}Q_{ik}$, and $F_{ik} = P_{ik}C_{ik}$. The producer revenue in sector k of country i is

$$R_{ik} = \sum_{j=1}^{4} \pi_{jik} X_{jk}. \tag{14}$$

So, Equation (13) can be written as:

$$X_{ik} = F_{ik} + \sum_{n\in\{a,m,s\}} \gamma_{ink}\lambda_{in}R_{in}. \tag{15}$$

In particular, the intermediate requirements of an exporting industry depend on its sales to all destinations.

3.4 Trade Policy and Structural Change

In this section, I examine the theoretical drivers of structural change, focusing on how trade integration and development characteristics of trading partners interact with these forces. To facilitate analytical tractability and yield transparent insights, I adopt a simplified version of the model. Specifically, I assume (i) no intermediate inputs, i.e., $\lambda_{ik} = 0$, so production depends only on labor and idiosyncratic productivity, and (ii) balanced trade in every period, i.e., $\phi_i = 0$, so that income equals total expenditure in each country. For this analytical illustration, I additionally impose equal sectoral wages, $\xi_{ik} = 1$. The quantitative model retains the calibrated sectoral wage wedges.

Let $\ell_{ik} := L_{ik}/L_i$ denote the share of labor allocated to sector k in country i, which is also equal to the value-added share in this simplified framework. I aim to derive an explicit expression for ℓ_{ik} under these simplifying assumptions. Starting from the equilibrium condition for labor market clearing $(S4)$, the labor share can be expressed as:

$$\ell_{ik} = \frac{L_{ik}}{L_i} = \sum_{j=1}^{4} \frac{P_{jk}Q_{jk}\pi_{jik}}{w_i L_i} = \sum_{j=1}^{4} \frac{P_{jk}Q_{jk}}{w_j L_j} \frac{w_j L_j}{w_i L_i} \pi_{jik} \tag{16}$$

Given $\lambda_{ik} = 0$, this implies that labor is the only input into production and that intermediate input usage is zero. Therefore, Equation (15) implies that total output in sector k of country j equals final consumption: $P_{jk}Q_{jk} = P_{jk}C_{jk}$. Substituting into Equation (16), the expression simplifies to:

$$\ell_{ik} = \sum_{j=1}^{4} e_{jk} \cdot \frac{w_j L_j}{w_i L_i} \cdot \pi_{jik}, \tag{17}$$

where $e_{jk} := P_{jk}C_{jk}/E_j$ is the share of expenditure on sector k in country j, and E_j is total household expenditure. To derive e_{jk}, I use the sectoral demand expression from Equation (10) and the price index in Equation (11), which together yield:

$$e_{jk} = \omega_{jk}^{\sigma} \left(\frac{P_{jk}}{P_j}\right)^{1-\sigma} \left(\frac{C_j}{L_j}\right)^{(1-\sigma)(\varepsilon_k - 1)}. \tag{18}$$

Using the price index identity from Equation (11), I obtain an expression for the sectoral price ratio $(P_{ik}/P_i)^{1-\sigma}$ as:

$$\left(\frac{P_{ik}}{P_i}\right)^{1-\sigma} = \left[\sum_{l \in \{a,m,s\}} \omega_{il}^{\sigma} \left(\frac{P_{il}}{P_{ik}}\right)^{1-\sigma} \left(\frac{C_i}{L_i}\right)^{(1-\sigma)(\varepsilon_l - 1)}\right]^{-1}. \tag{19}$$

Substituting (19) into (18), the expression for e_{ik} becomes:

$$e_{ik} = \left[\sum_{l \in \{a,m,s\}} \left(\frac{\omega_{il}}{\omega_{ik}}\right)^{\sigma} \left(\frac{P_{il}}{P_{ik}}\right)^{1-\sigma} \left(\frac{C_i}{L_i}\right)^{(1-\sigma)(\varepsilon_l - \varepsilon_k)}\right]^{-1}. \tag{20}$$

Finally, the share of import of country j for sector-k goods sourced from sector i, π_{jik}, can be rewritten as:

$$\pi_{jik} = \left[1 + \sum_{l \neq i} \frac{T_{lk}}{T_{ik}} \left(\frac{v_{lk}}{v_{ik}}\right)^{-\theta} \left(\frac{\tau_{jlk}}{\tau_{jik}}\right)^{-\theta}\right]^{-1}. \tag{21}$$

Equation (20) illustrates how the two traditional mechanisms of structural change shape the sectoral employment shares. The first is the Engel effect, which arises from the nonhomotheticity of preferences, captured here by sector-specific income elasticities of demand. As aggregate consumption increases with income, expenditure shifts toward sectors with higher income elasticities. Consequently, the expenditure and employment shares of such sectors rise, while sectors with lower income elasticities experience a decline in both. This shift in labor shares is formalized in Equation (17), where changes in expenditure share e_{ik} induced by income growth translate into sectoral labor share ℓ_{ik}.

The second mechanism is the Baumol effect, arising from differential sectoral productivity growth and non-unitary substitution across sectoral goods. With $\sigma < 1$, a relative-price decline lowers the sector's expenditure share conditional on the consumption index: the quantity response is less than proportional to the price decline. Conversely, a relative-price increase raises its expenditure share under the same condition. Sectoral employment also depends on sourcing shares and equilibrium income and wages, so the expenditure response alone does not determine the overall reallocation.

Equations (17) (20) and (21) together illustrate how trade cost reduction due to trade integration affects structural change by modifying both the relative prices of sectoral goods and the pattern of bilateral trade shares. First, a reduction in trade costs in sector k lowers the price index P_{ik} relative to other sectors. Given non-unitary substitution elasticities ($\sigma \neq 1$), this

relative price decline reduces the sector's expenditure share e_{ik} via Equation (20), leading to a reallocation of labor away from sector k. Second, as shown in Equation (21), a decline in bilateral trade costs τ_{jik} for varieties produced in country i and exported to country j enhances the competitiveness of country i's goods in international markets. This increase in competitiveness raises the trade share π_{jik} in Equation (21), reflecting higher external demand for sector k's goods from country i. The resulting expansion in trade shares generates upward pressure on labor demand in sector k, potentially offsetting the decline induced by the relative price effect.

Moreover, Equation (21) provides a direct link between partner development and the sectoral effects of trade-cost reductions. A decline in the bilateral trade cost τ_{jik} increases country i 's competitiveness in country j 's market and, all else equal, raises its share π_{jik} of country j 's expenditure on sector- k goods. The magnitude and sectoral composition of the resulting trade flows, however, depend on the characteristics of the economies connected by the reduction in trade costs. In particular, country i 's competitiveness in sector k depends on its fundamental productivity T_{ik} and its unit production cost, v_{ik}.

These objects capture several dimensions along which economies at different stages of development may differ, including sectoral productivity, labor costs, intermediate-input prices, and production structure. A higher T_{ik}, holding competitors' technologies fixed, raises π_{jik}, while lower production costs strengthen country i 's competitiveness in sector k. What matters for trade-induced structural change is therefore not simply the aggregate productivity or income level of a trading partner, but its sectoral configuration of productivity and production costs, which determines its pattern of comparative advantage.

The development level of trading partners also matters through the demand side. As Equations(17) shows, the contribution of foreign demand to employment in sector k depends not only on the trade share π_{jik}, but also on the partner's sectoral expenditure share e_{jk} and relative market size $w_j L_j / (w_i L_i)$. Because nonhomothetic preferences generate systematic changes in expenditure patterns with income, economies at different levels of development demand different compositions of sectoral goods. A reduction in trade costs with an advanced economy can therefore generate a different pattern of external demand from an otherwise identical reduction in trade costs with a less developed economy. Moreover, a larger partner market magnifies the extent to which these demand patterns affect sectoral labor demand in the home economy.

Partner development therefore interacts with trade integration through both supply and demand channels. On the supply side, sectoral productivity, wages, and intermediate-input costs determine relative competitiveness and hence bilateral trade shares. On the demand side, income and expenditure patterns determine the sectoral composition of the foreign demand to which lower trade costs provide greater access. These forces need not operate in the same direction. For example, integration with an advanced economy may expose domestic producers to strong foreign competition in sectors in which the partner has a comparative advantage, while simultaneously increasing foreign demand for sectors in which the home economy is relatively competitive. The resulting sectoral reallocation depends on the balance between these forces.

Taken together, Equations (21), (17), and (20) imply that the effects of a given trade-cost reduction on structural change depend on the development characteristics of the economies it connects. Identical changes in trade costs can generate different sectoral trade flows, expenditure patterns, and labor allocations when trading partners differ in productivity, production costs, income, expenditure composition, or market size. For Mexico, this mechanism provides a theoretical rationale for why integration with the more advanced United States and Canada may have generated a different pattern of structural change than integration with economies at Mexico's own level of development. Because the different channels can reinforce or offset one another, however, the direction and magnitude of the overall effect cannot generally be determined analytically. I therefore turn next to the quantitative analysis, beginning with the calibration of the model.

4 Calibration

This section describes the strategy used to discipline the model with data. The parameters fall into two groups: time-invariant structural parameters that are either set from the literature or identified from cross-sectional moments, and exogenous time series that are recovered period by period to match observed data. Before describing the calibration procedure, I first describe the data employed in the analysis and their sources.

4.1 Data Description

The model contains four countries: Mexico, the United States, Canada, and a rest-of-the-world (ROW) which includes Australia, Austria, Belgium, Brazil, China, Germany, Denmark, Spain, Finland, France, United Kingdom, Greece, India, Ireland, Italy, Japan, South Korea, Luxembourg, Netherlands, Portugal, Sweden and Taiwan. The dataset covers the period 1971–2018. Refer to Appendix C for data details and coverage notes.

I aggregate data into three sectors following the International Standard Industrial Classification of All Economic Activities (ISIC), Revision 4: agriculture (including forestry and fishing), manufacturing (a broad industrial aggregate including mining, utilities, and construction), and services (all remaining activities). The calibration requires sectoral data for employment, value added, gross output, input-output coefficients, imports, exports, and gross-output prices. These data come from several sources.

Sectoral employment, nominal and real value added, real and nominal gross output, bilateral trade flows, and input-output linkages are compiled primarily from EU KLEMS, WIOD, and the OECD Inter-Country Input-Output (ICIO) tables. For countries or years not covered by those sources, the series are supplemented with the GGDC 10-Sector Database, the GGDC Productivity Level Database, BEA employment data for the United States, Statistics Canada employment data, UN sectoral value-added data for selected gaps, Penn World Table population and income data, and World Bank WDI services-trade data. Sectoral gross-output price indexes $P_{ik,t}$, used throughout the calibration as the price of the composite good in Equation (3), are constructed as nominal gross output divided by PPP-adjusted real gross output. Bilateral trade flows, measured in millions of U.S. dollars, are used to construct the bilateral trade share matrices $\pi_{ijk,t}$, while the input-output coefficients λ_{ik} and γ_{ikn} are calibrated directly from input-output tables and averaged over the available sample period.

For ROW variables, employment and nominal variables are aggregated by summation. Real variables are constructed using Tornqvist growth aggregation within each source and then benchmarked to 2005 PPP prices.[3]

Having described the data, I now turn to the calibration procedure. I begin by calibrating the parameters that are assumed to remain constant over time.

4.2 Time-Invariant Parameters

The time-invariant parameters of the model include the trade elasticity θ, the elasticity of substitution across varieties η, the preference parameters σ, ε_k, and ω_{ik}, as well as the production parameters λ_{ik} and γ_{ikn}.

I set the trade elasticity to $\theta = 4$. This value lies within the range of empirical estimates surveyed by Simonovska & Waugh (2014) and is widely adopted in quantitative trade models of structural change (Uy et al. 2013, Sposi 2019). I also set the elasticity of substitution across varieties to $\eta = 2$, following De Vries et al. (2026). As in Eaton–Kortum models, this parameter plays only a limited quantitative role in determining equilibrium allocations.[4]

[3] Appendix C.7 describes the Tornqvist aggregation procedure.

[4] The chosen value satisfies the restrictions $\eta > 1$ and $\eta < 1 + \theta$, ensuring that the Frechet price index remains well defined.

For preferences, I adopt the estimates in Sposi et al. (2026), which are estimated on the basis of a sample of 28, comprising the 25 countries included in this study and three other countries. Specifically, I set the elasticity of substitution across sectoral composite goods to $\sigma = 0.233$, and the Engel-curve parameters to $\varepsilon_a = 0.086$, $\varepsilon_m = 0.860$, and $\varepsilon_s = 1.146$.

The remaining preference parameters are the sectoral preference weights ω_{ik}. I calibrate these parameters separately for each country by choosing the values that best replicate, in the initial year of the sample, both the observed sectoral labor shares and the cross-country gap in real consumption per worker relative to Mexico.[5]

Finally, I calibrate the production parameters λ_{ik} and γ_{ikn} using input–output data. The parameter λ_{ik} is computed as the average intermediate-input share in nominal gross output, $1 - VA_{ik}/GO_{ik}$, for sector k in country i. The input–output coefficients γ_{ikn}, which measure the sectoral composition of intermediate inputs used in sector k, are constructed from national input–output tables after aggregating industries into the three sectors of the model. For each country, I compute these shares annually and use their time average in the calibration. Table 1 reports values of key parameters yielded by calibration.

Table 1: Time-Invariant Parameters

Description	Param.	Value	Source
Trade elasticity	θ	4	Simonovska & Waugh (2014)
Substitution between varieties	η	2	De Vries et al. (2026)
Substitution between sectors	σ	0.233	Sposi et al. (2026)
Engel-curve parameters	ε_a	0.086	Sposi et al. (2026)
	ε_m	0.860	Sposi et al. (2026)
	ε_s	1.146	Sposi et al. (2026)

Ratio of intermediates to gross output	Agr. λ_{ia}	Mfg. λ_{im}	Svc. λ_{is}
Mexico	0.330	0.576	0.228
Canada	0.547	0.600	0.352
United States	0.564	0.581	0.349
Rest of World	0.416	0.662	0.381

Notes: The table reports time-invariant parameters used in the baseline calibration. Preference parameters are taken from Sposi et al. (2026); production parameters are calibrated from the input-output data as described in Section 4.2.

4.3 Time-Varying Exogenous Processes

This subsection describes the calibration of the main time-varying exogenous variables in the model: wage wedges, trade imbalances, sectoral productivity and bilateral trade costs. Each country's labor endowment is set equal to observed total sectoral employment.

The model allows for sectoral wage differentials within each country and calibrates them from the data. The wage in sector k of country i satisfies $w_{ik} = \xi_{ik} w_{im}$, where the manufacturing wedge ξ_{im} is normalized to one. I compute the wedge directly as

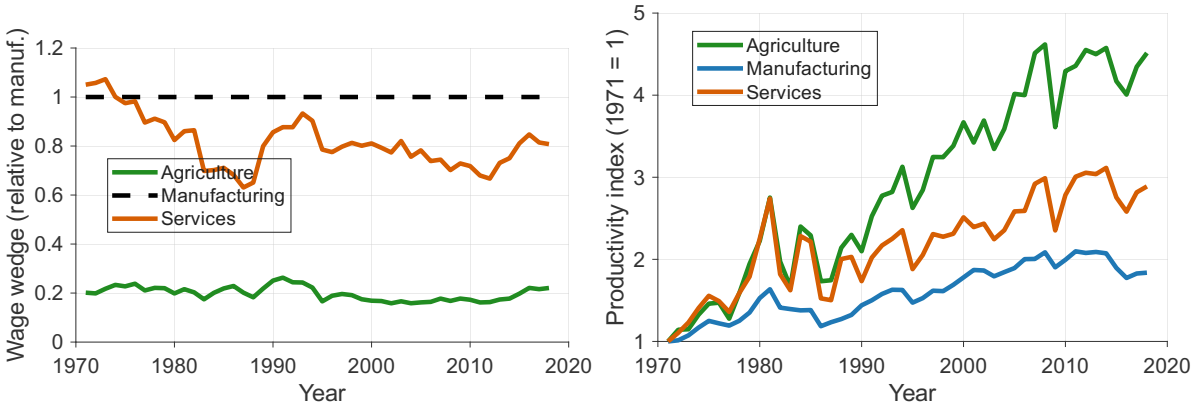
$$\xi_{ik,t} = \frac{VA_{ik,t}/L_{ik,t}}{VA_{im,t}/L_{im,t}}, \quad k \in \{a,\, s\} \tag{22}$$

These ratios are computed from the data in each year and treated as time-varying observed parameters rather than structural parameters to be solved for. This approach follows Buera &

[5] I target cross-country income gaps because development differences play a central role in the analysis.

Kaboski (2009) and Swiecki (2017), who argue that observed wage differentials reflect labor-market frictions and distortions that are better modeled as wedges than as equilibrium outcomes.

The left panel of Figure 3 shows that these wedges are quantitatively important for Mexico. Agricultural value added per worker is persistently far below manufacturing value added per worker: the agricultural wedge is 0.20 in 1971, fluctuates between 0.16 and 0.26, and is 0.22 in 2018. The services wedge is closer to manufacturing, but it declines from 1.05 in 1971 to 0.81 in 2018. These patterns matter for the model because employment and value-added shares need not move one-for-one in the data. If all sectoral wages were forced to be equal, the model would attribute too much of the gap between employment and value added to technology or trade costs. Treating $\xi_{ik,t}$ as an observed wedge allows the calibration to preserve measured sectoral labor-cost differences while focusing the counterfactual analysis on technology, trade costs, and partner characteristics.



(a) Mexico: Sectoral Wage wedges

(b) Mexico: Calibrated Sectoral Productivities

Figure 3: Mexico: Calibrated Sectoral Wage Wedges and Productivities

Notes: The left panel plots calibrated sectoral wage wedges $\xi_{ik,t}$ for Mexico. The manufacturing wedge $\xi_{im,t}$ is normalized to one. The right panel plots calibrated sectoral productivity indexes, normalized to the 1971 value.

The transfer coefficients $\phi_{i,t}$ are recovered from observed net exports and the global portfolio accounting identity, with the ROW coefficient normalized to zero. The common transfer is determined by condition (8). Counterfactuals preserve these coefficients, while the transfer and net-export levels respond to equilibrium incomes.

I compute productivity in two steps. I retain the direct inversion of the domestic trade-share equation:[6]

$$T_{ik,t} = \pi_{iik,t} \left(\frac{P_{ik,t}}{G} \right)^{-\theta} v_{ik,t}^{\theta}, \tag{23}$$

where G is the Fréchet price parameter and $v_{ik,t}$ the unit costs. In the calibration, observed value added per worker is used to construct sectoral wage wedges and the empirical unit costs that recover $T_{ik,t}$ and $\tau_{jik,t}$.

Taking the ratio of an import share to the importer's domestic share in Equation (5) eliminates the common price-denominator term:

$$\frac{\pi_{jik,t}}{\pi_{jjk,t}} = \frac{T_{ik,t}\, v_{ik,t}^{-\theta}\, \tau_{jik,t}^{-\theta}}{T_{jk,t}\, v_{jk,t}^{-\theta}},$$

[6]This relation yields from a combination of conditions S1 and S3 in Table A.5 so that the baseline calibrations are designed to map the observed sectoral prices, $P_{ik,t}$, and absorption shares, $\pi_{iik,t}$.

and therefore

$$\tau_{jik,t} = \left[\frac{T_{ik,t}}{T_{jk,t}} \left(\frac{v_{ik,t}}{v_{jk,t}} \right)^{-\theta} \frac{\pi_{jjk,t}}{\pi_{jik,t}} \right]^{1/\theta}, \qquad i \neq j, \tag{24}$$

with $\tau_{iik,t} = 1$.

The right panel of Figure 3 reports the implied productivity paths for Mexico, expressed as the Fréchet productivity index $A_{ik,t} \propto T_{ik,t}^{1/\theta}$, normalized to 1971. It shows that productivity growth is sector biased. By 2018, Mexico's productivity index is 4.51 in agriculture, 1.84 in manufacturing, and 2.89 in services, relative to 1971. Table A.1 in Appendix A.1 reports the same productivity index for all countries in the model while Figure A.1 shows their evolution. The same pattern emerges across countries: agriculture has the highest growth, followed by services and then manufacturing.

Following the interpretation in De Vries et al. (2026), I view the recovered $\tau_{jik,t}$ series as broad bilateral-sectoral wedges rather than tariff measures alone. They summarize the combination of tariffs, transportation costs, border frictions, non-tariff barriers, measurement error, and other forces needed for the model to rationalize observed bilateral sourcing. Figure 4 shows that these wedges are highly asymmetric across partners and sectors. U.S.-to-Mexico costs are lower than Canada-to-Mexico costs throughout most of the sample and decline in agriculture and manufacturing between 1971 and 2018, from 3.89 to 2.33 and from 2.07 to 1.58, respectively. Canada-to-Mexico costs begin at much higher levels, especially in agriculture, and fall from 12.29 to 3.00 in agriculture, 4.79 to 3.22 in manufacturing, and 5.41 to 3.71 in services.

The outbound paths reinforce the same point. Mexico-to-U.S. costs fall between 1971 and 2018 from 3.24 to 1.96 in agriculture, 5.68 to 2.31 in manufacturing, and 8.40 to 5.74 in services. Mexico-to-Canada costs fall from 3.02 to 2.73 in agriculture, 6.67 to 2.45 in manufacturing, and 11.45 to 6.09 in services. Services remain the highest-cost sector for export from Mexico to both North American destinations, with high values between the 1980s and 1990s.

I now turn to the model's fit to the main allocation moments.

4.4 Model Fit

I assess the model along two dimensions that are central to the analysis: sectoral employment shares and nominal value-added shares. This distinction is particularly important for a country such as Mexico, where sectoral employment and value-added shares differ substantially. Examining both outcomes allows me to assess how economic policies affect the allocation of labor and economic activity across sectors, as well as whether they contribute to a convergence between these two dimensions of structural change.

The calibration matches the initial sectoral allocation through the preference weights and uses observed prices and trade shares to recover the paths of sectoral productivity and bilateral trade costs. Importantly, the calibration does not mechanically target the subsequent paths of employment or value-added shares. The model-fit exercise therefore provides an informative assessment of whether the calibrated economy can reproduce the broad evolution of structural change using a parsimonious set of preference parameters, sectoral wage wedges, productivity paths, and bilateral trade costs.

Table 2 summarizes the fit over 1971–2018. Across country-sector cells, the employment-share RMSE is 2.20 percentage points and the value-added-share RMSE is 1.86 points. For Mexico, the sectoral employment RMSEs are 3.25 points in agriculture, 2.86 in manufacturing, and 2.90 in services. The corresponding value-added errors are 1.06, 3.33, and 3.23 points. Mexico's manufacturing employment correlation is 0.46, while the RMSE in that sector is moderate (2.86 percentage points), indicating that the model is unable to capture the inflection points observed between 1980 and 1990. As can be seen in the figure, the fit remains very good.

Figures 5 and 6 provide the corresponding visual comparison for Mexico and its partners. For Mexico, the model captures the decline in agriculture and the rise of services, while the

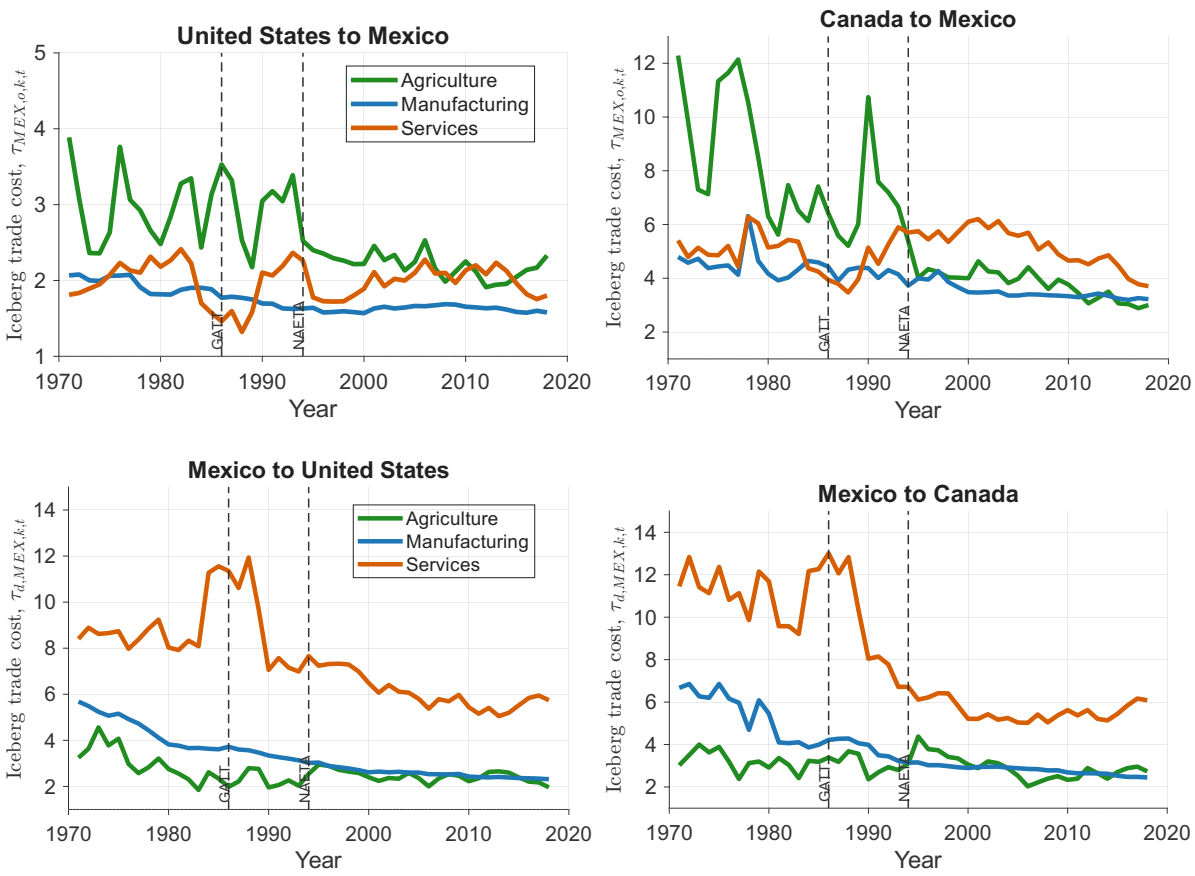


Figure 4: Mexico: Calibrated Inbound and Outbound Trade Costs

Notes: The top row plots calibrated iceberg trade costs $\tau_{MEX,o,k,t}$ for goods shipped from origin $o \in \{USA, CAN\}$ to Mexico. The bottom row plots $\tau_{d,MEX,k,t}$ for Mexican shipments to destination $d \in \{USA, CAN\}$. Vertical lines mark GATT entry in 1986 and NAFTA in 1994.

manufacturing path is harder to match year by year. For the partner economies, the fit is smoother, reflecting the more gradual sectoral transitions in Canada, the United States, and the ROW aggregate. The main implication is that the model reproduces the level and direction of the allocation moments sufficiently well to use it for comparative counterfactuals.

Figure A.4 and Table A.2 in Appendix A provide an additional diagnostic based on nominal gross-output shares. These moments are not direct allocation targets in the calibration, which is disciplined primarily by employment, value added, prices, input-output coefficients, and trade shares. Across country-sector cells, the RMSE is 2.94 percentage points and the mean time-series correlation is 0.855. Therefore, the model also tracks the sectoral composition of gross output supports the internal consistency of the production side of the calibrated economy.

Overall, our calibration reproduces the most relevant empirical patterns of structural change for the four countries over time. I now turn to counterfactual exercises that isolate how the development level of trade partners affects Mexico's structural-change path.

5 Quantitative Counterfactual Experiments

In this section, I quantify how trading partners' development characteristics shape the effects of trade integration on Mexico's structural change. Although the existing literature has quantified the effects of trade-cost shocks on structural change, these effects are conditional on the characteristics of the economies with which a country trades. In particular, identical reductions in

Table 2: Model Fit: Employment and Value-Added Shares

	RMSE (pp)			Correlation		
Country	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
Panel A: Employment shares						
Mexico	3.25	2.86	2.90	0.943	0.465	0.959
Canada	0.71	1.35	1.94	0.885	0.921	0.925
United States	0.28	1.46	1.64	0.885	0.995	0.995
Rest of World	3.40	1.05	2.71	0.945	0.932	0.942
All countries		2.20			0.899	
Panel B: Nominal value-added shares						
Mexico	1.06	3.33	3.23	0.959	0.807	0.924
Canada	0.55	1.62	2.05	0.950	0.863	0.888
United States	0.24	1.79	1.91	0.980	0.989	0.991
Rest of World	0.67	1.22	1.74	0.948	0.948	0.952
All countries		1.86			0.933	

Notes: RMSE is the root mean squared error in percentage points between model and data shares over 1971–2018. Model nominal value added is the sectoral labor bill $w_{ik}L_{ik}$. Correlation is the time-series Pearson correlation between simulated and observed shares.

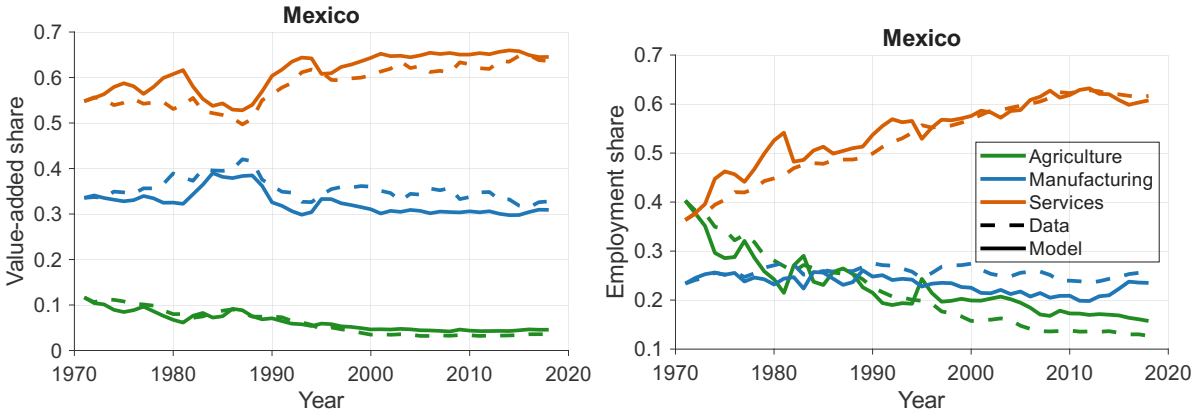
trade costs may generate different patterns of sectoral reallocation depending on trading partners' income levels, expenditure patterns, and production structures. To identify this dimension, I conduct a series of counterfactual experiments that hold trade-cost changes fixed while varying partners' development characteristics, thereby isolating their role in shaping Mexico's structural response to trade integration.

5.1 Experiment Design

I construct counterfactual environments to isolate how differences in development levels between trading partners shape the effects of trade integration on structural change. In this environment, sectoral wage wedges, preference weights and the initial sectoral productivity levels of the United States and Canada with their Mexican counterparts. I treat these three components as development characteristics because they summarize the productive efficiency level, sectoral labor-cost structure, relative earnings, and initial expenditure composition that differ systematically between middle-income and high-income economies.

The productivity replacement is an initial-level shift: each partner keeps its own sector-specific productivity-growth path after 1971. I include preference weights in the development bundle because they are calibrated to match country-specific initial expenditure shares, which vary systematically with the level of development. The resulting environment approximates a setting in which Mexico trades with partners at a similar stage of development. I refer to this environment as *Mexico-symmetric*. The partners do not become complete replicas of Mexico. Their labor endowments, input-output coefficients, transfer coefficients, bilateral trade-cost paths, and technology-growth paths retain their baseline values. Wages, expenditure, prices, trade shares, and net-export levels are solved endogenously.

The counterfactual replacement of initial sectoral productivity levels, sectoral wage wedges, and preference weights is not an arbitrary relabeling of countries. Figures A.2 and A.3 in Appendix A.2 plot the calibrated sectoral wedges and proclivities for Mexico, Canada, the United States, and the rest of the world. These plots reveal systematic differences associated with countries' levels of development. Initial sectoral productivity levels are relatively close in Canada and the United States but differ substantially from those in Mexico; sectoral wage wedges also differ across development levels. Moreover, preference weights are calibrated to



(a) Sectoral Value-Added Shares (b) Sectoral Employment Shares

Figure 5: Sectoral Value-Added and Employment Shares: Data and Model

Notes: The left panel plots Mexico's nominal value-added shares and the right panel plots Mexico's employment shares. Dashed lines are observed data and solid lines are model-implied shares from the baseline annual calibration. Colors identify agriculture, manufacturing, and services.

match country-specific initial expenditure shares, which vary systematically with the level of development. Accordingly, preference weights summarize the country-specific demand systems recovered from initial employment allocations and income gaps across countries.

I evaluate the effects of global trade integration and two major trade liberalization episodes, GATT accession and NAFTA, in both the baseline and *Mexico-symmetric* environments. Specially, *no-integration* counterfactual holds bilateral trade costs involving Mexico fixed at their 1971 levels throughout the sample. The *no-GATT* counterfactual holds Mexico's bilateral trade costs at their 1985 levels from 1986 onward, while the *no-NAFTA* counterfactual holds intra-North-American trade costs at their 1993 levels from 1994 onward. Applying these experiments in the *Mexico-symmetric* environment allows me to assess how Mexico's structural change would have responded to the same trade-cost reductions had the United States and Canada shared Mexico's development characteristics rather than those of advanced economies. Comparing the effects of trade cost reduction across baseline and *Mexico-symmetric* environments isolates the contribution of differences in development levels between Mexico and its trading partners to trade-induced structural change.

One construction detail deserves emphasis. The *no-GATT* counterfactual freezes Mexico's bilateral trade costs at their 1985 level from 1986 onward, through the end of the sample in 2018; it is not reactivated in 1994. Consequently, the reported GATT effect is not restricted to the 1986–1993 window and mechanically nests the subsequent NAFTA-era trade-cost decline within it. The GATT and NAFTA effects reported below should therefore be read, respectively, as the effect of Mexico's entire post-1985 trade-cost path (inclusive of NAFTA) and as the effect of the incremental North-American-specific trade-cost change after 1993, rather than as two mutually exclusive components of an additive decomposition of total trade integration. I adopt this convention because it corresponds to the natural counterfactual question of what would have happened had Mexico's post-1985 opening never occurred, while ensuring that the GATT- and NAFTA-specific magnitudes are not misread as summing to the total integration effect.

The calibrated exogenous series exhibit substantial year-to-year variation in some periods. Because the model is solved as a sequence of static annual equilibria, annual fluctuations in technologies, trade costs, and sectoral wage wedges pass directly into the counterfactual outcomes.[7] The resulting fluctuations should therefore be interpreted as equilibrium responses to high-frequency variation in the recovered exogenous environment. To distinguish these move-

[7]These fluctuations do not reflect numerical instability. In diagnostic checks, the complete set of counterfactual equilibrium converges at the specified solver tolerance, with negligible equilibrium residuals

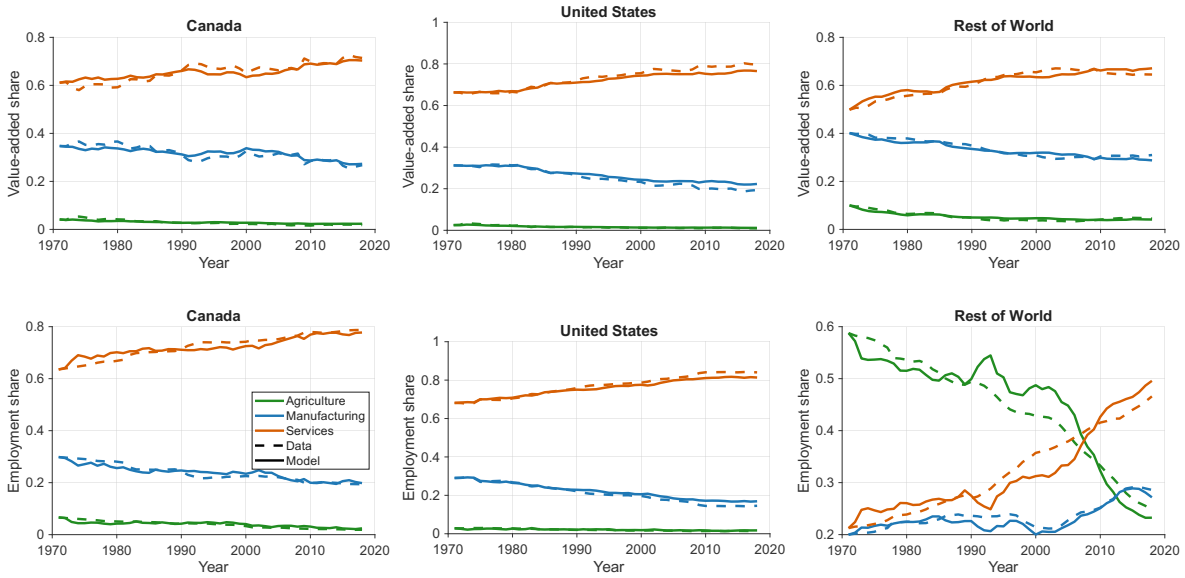


Figure 6: Sectoral Value-Added and Employment Shares: Data and Model, Partner Economies

Notes: The first row plots nominal value-added shares and the second row plots employment shares for Canada, the United States, and the rest of the world. Dashed lines are observed data and solid lines are model-implied shares from the baseline annual calibration. Colors identify agriculture, manufacturing, and services.

ments from the lower-frequency economic patterns of interest, I report results from both the annual calibration and an additional set of simulations based on smoothed exogenous primitives.

For the smoothed specification, I conduct a second set of simulations using HP-filtered exogenous primitives.[8] Specifically, I filter sector-country technologies, sectoral wage wedges, and bilateral trade costs. I filter trade costs separately over 1971–1985, 1986–1993, and 1994–2018 to prevent the filtering procedure from mechanically shifting trade-cost changes associated with GATT or NAFTA across policy regimes. Labor endowments, transfer-rule coefficients, preference weights, and input–output coefficients remain at their calibrated values. Importantly, the smoothing is applied only to the exogenous primitives. I then recompute the equilibrium using these filtered series, rather than smoothing the resulting equilibrium outcomes ex post.

The quantitative object of interest is the interaction between trade integration and partner characteristics. Let $\mathcal{D}$ denote a partner environment and let $s^q_{k,t}(\tau, \mathcal{D})$ be Mexico's share for outcome $q \in \{L, VA\}$. Define the integration effect as

$$I^q_{k,t}(\mathcal{D}) = 100 \left[s^q_{k,t}(\tau^O, \mathcal{D}) - s^q_{k,t}(\tau^F, \mathcal{D}) \right]. \tag{25}$$

where τ^O and τ^O are the calibrated and frozen trade costs, respectively. For the actual partner environment $\mathcal{D}^A$ and the *Mexico-symmetric* environment $\mathcal{D}^M$, the interaction is

$$\mathcal{J}^q_{k,t} = I^q_{k,t}(\mathcal{D}^A) - I^q_{k,t}(\mathcal{D}^M). \tag{26}$$

This comparison uses four equilibria. It differs from the difference between sectoral shares under the two partner environments at observed trade costs, which also contains differences that would arise under the frozen-cost counterfactual. The distinction is central to interpreting the results below.

[8] I apply the Hodrick–Prescott filter to strictly positive primitives in logs, using $\lambda = 6.25$, as recommended by Ravn & Uhlig (2002) for annual data.

5.2 Development Level of the Trade Partner and Structural Change

To quantify the role of trading partners' development characteristics in shaping Mexico's structural change, this section proceeds in four steps. First, I examine how Mexico's structural change would have differed had its North American trading partners shared its level of development. Second, I compare the effects of trade-cost reductions on structural change between the baseline and a counterfactual in which Mexico's trading partners have similar development characteristics. Third, I decompose the differences in these effects to quantify the contribution of each component of the development-characteristics bundle. Finally, I conduct additional experiments distinguish the common level of partner technology from its sectoral composition.

5.2.1 Absolute Role of Development Level of the Trade Partner

Figure 7 compares Mexico's predicted employment and nominal value-added shares under the baseline simulation with those obtained in the *Mexico-symmetric* environment. The lighter lines report results based on the annually calibrated primitives, whereas the darker lines report equilibrium outcomes recomputed using HP-filtered primitives. The two specifications deliver similar lower-frequency patterns, indicating that the main results are not driven by the high-frequency variation in the annually calibrated exogenous series.

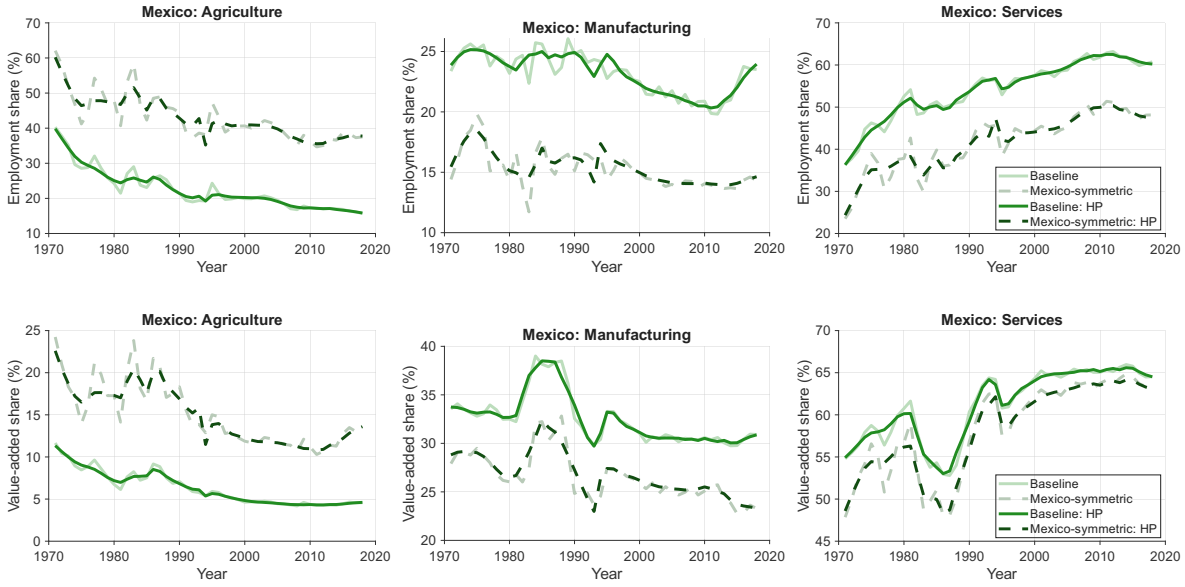


Figure 7: Structural Change for Actual Mexico and *Mexico-Symmetric*

Notes: The figure compares the baseline predictions with *Mexico-symmetric* predictions, in which the United States and Canada receive Mexico-level initial productivity levels, sectoral wage wedges, and preference weights while retaining their own productivity-growth paths. The top panels plot sectoral employment shares while the bottom panels plot nominal value-added shares. Lighter lines use the annual calibrated primitives; darker lines use equilibria recomputed from HP-filtered primitives.

The results show that Mexico's pattern of structural change would have differed substantially had its trading partners shared its development characteristics. On average, the agricultural employment share in the baseline is approximately 20 percentage points lower than in the Mexico-level partners environment, while the manufacturing and services shares are 9 and 11 percentage points higher, respectively. These differences reflect the role of partner development in shaping sectoral allocations under observed trade costs. They should, however, be distinguished from differences in the effects of trade integration, which are measured by comparing sectoral allocations under observed and counterfactual frozen trade-cost paths within each environment.

The findings reveal that trading with advanced economies contributes to a faster movement of workers out of agriculture. This movement out of agriculture relates to Tombe (2015), who studies how agricultural imports could naturally help alleviate the problem of high employment in low productivity agriculture, and to Teignier (2018), who shows how agricultural imports can facilitate structural transformation in Great Britain and South Korea. The experiment here isolates a complementary margin: the role of partner development characteristics at a given bilateral trade-cost path.

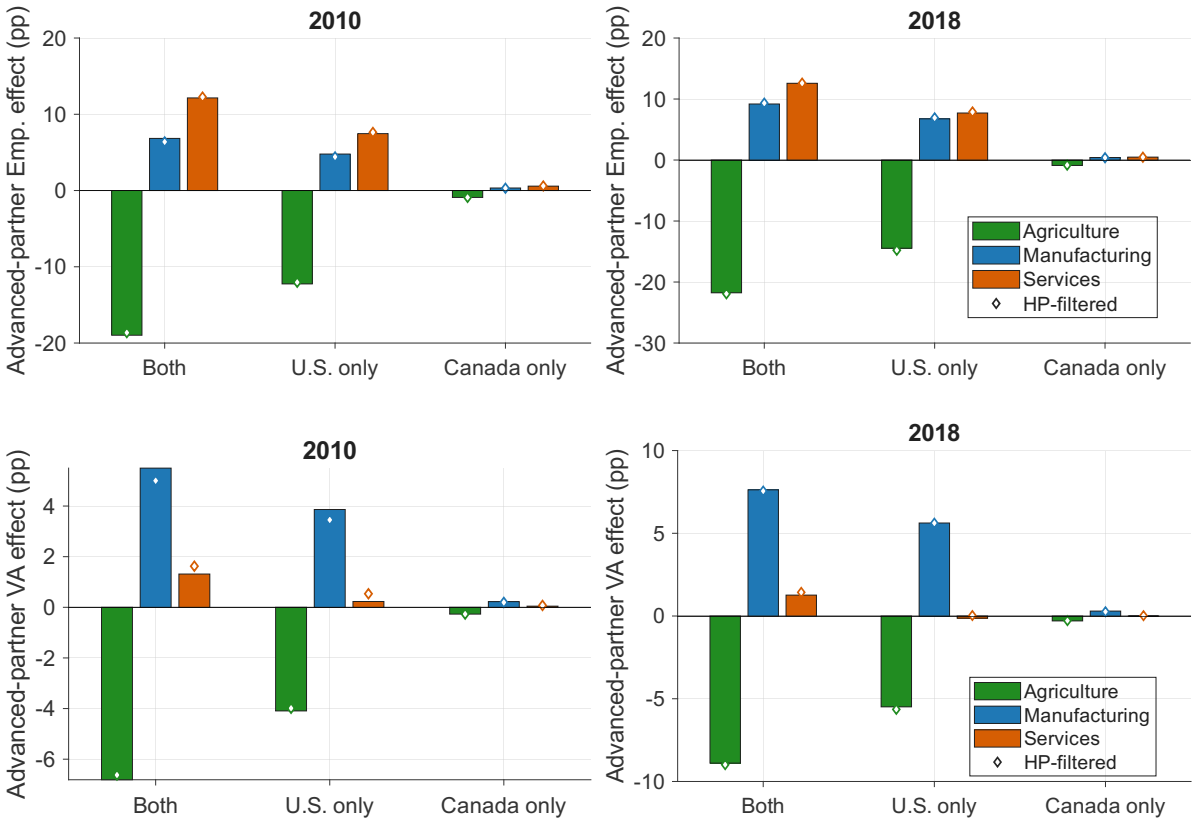


Figure 8: Mexico: Partner Contributions to the Development Effect

Notes: The four panels report Mexico's advanced-partner effects in 2010 and 2018. The top row plots effects on employment share, and the bottom row plots effects on nominal value-added share. Within each panel, bars compare the effects of replacing the development characteristics of both North American partners, the United States only, or Canada only with Mexico-level counterparts. Positive bars mean that actual advanced partners raise Mexico's sector share relative to the corresponding Mexico-level-partner environment. Bars use annual calibrated primitives; diamond markers report equilibria recomputed from HP-filtered primitives.

The results for nominal value-added shares indicate that integration with more advanced trading partners is associated with a reallocation of economic activity from agriculture toward manufacturing. Relative to the *Mexico-symmetric*, the baseline agricultural value-added share is approximately 6 percentage points lower on average over the sample period. These findings suggest that the development characteristics of Mexico's trading partners play an economically important role in shaping both the employment and value-added shares.[9]

To identify which trading partner drives the partner-development effect, I replace U.S. and

[9]The difference between the employment and value-added responses is relevant in light of the large gaps in value added per worker between agriculture and nonagriculture documented by Gollin et al. (2014), especially in developing countries. Employment shares and nominal value-added shares consequently capture distinct dimensions of structural change; the productivity-gap evidence alone does not identify the labor-market wedges used in this model.

Canadian development characteristics separately. Figure 8 and Table A.3 in Appendix A.4 report results. These results show that assigning Mexico-level development characteristics to the United States alone generates employment and value-added shares that are close to those obtained when both the United States and Canada are reassigned Mexico-level development characteristics. In 2018, baseline employment shares minus those with a Mexico-level United States are $(-14.46, +6.75, +7.71)$ percentage points in agriculture, manufacturing, and services. The corresponding Canada-only differences are $(-0.89, +0.41, +0.48)$ points. The joint comparison gives -21.77, $+9.19$, and $+12.58$ points, respectively. The separate U.S. replacement has the larger effect, but the two separate effects do not sum to the joint comparison due to their complementarity through general equilibrium effects.

5.2.2 Trade-Cost Changes and Partner Development

This section quantifies the contribution of Mexico's trade integration to structural change and examines how its effects depend on partner development. Table 3 and Figures 9 compares value added and employment shares predicted with observed versus frozen trade costs within baseline and *Mexico-symmetric* environments. In 2018, integration raises the baseline manufacturing value-added share by 8.49 percentage points and lowers the services share by 8.51 points, with an agricultural increase of 0.02 points. In the *Mexico-symmetric*, the corresponding effects are $+8.02$, $+1.88$, and -9.90 points for agriculture, manufacturing, and services. Manufacturing expands less while agriculture rises substantially more.

The employment-share results also depend on partner development, but differ from the nominal value-added results. Integration primarily reallocates labor from services toward manufacturing in the baseline, but toward agriculture under *Mexico-symmetric*. In the baseline framework, integration changes the agricultural, manufacturing, and services employment shares by $+0.28$, $+6.71$, and -6.99 percentage points in 2018 while its effects are $+19.47$, -1.35, and -18.13 points under the *Mexico-symmetric*. Thus, in 2018, the contribution of trading partners' development level to the effects of trade integration on employment shares is -19.19, $+8.06$, and $+11.14$ percentage points for agriculture, manufacturing, and services, respectively. The corresponding contributions to nominal value-added shares are -8.00, $+6.61$, and $+1.39$ percentage points. Since the partner replacement changes technology, wage wedges, and preferences jointly, this comparison alone does not identify the separate contribution of any one component. This decomposition will be provided later in the Section 5.2.3.

The policy-specific experiments reinforce these findings. The post-1985 trade-cost experiment (*no-GATT*) also depends on development characteristics. Baseline employment effects in 2018 are $(-6.78, +0.58, +6.20)$ percentage points in agriculture, manufacturing, and services. Under the *Mexico-symmetric* environment, they are $(+3.06, -3.27, +0.21)$ points. Figures A.5–A.7 report the annual paths. This change in magnitude and sign is also observed in the effect of post-1985 trade-cost experiment on sectoral shares of value added between the baseline and the *Mexico-symmetric* environments.

The *no-NAFTA* counterfactual examines the post-1993 change in intra-North-American trade costs. In 2018, the baseline effects on Mexico's nominal value-added shares are -0.77, $+2.13$, and -1.36 percentage points in agriculture, manufacturing, and services. Under the *Mexico-symmetric*, the effects are -0.81, $+0.75$, and $+0.07$ points. Moreover, the post-1993 trade-cost path raises manufacturing employment by 2.12 percentage points in the baseline and 0.70 points with Mexico-level partners. Thus, partner development amplifies the positive effect of NAFTA on manufacturing expansion.[10]

[10]These findings complement Caliendo & Parro (2015), who quantify NAFTA's trade and welfare effects in a Ricardian framework with intermediate inputs and sectoral linkages. My analysis focuses on employment and nominal value-added shares and on their dependence on partner development. The magnitudes are not directly comparable: their experiment evaluates tariff changes, whereas the experiment here freezes broader bilateral trade-cost wedges at their 1993 levels.

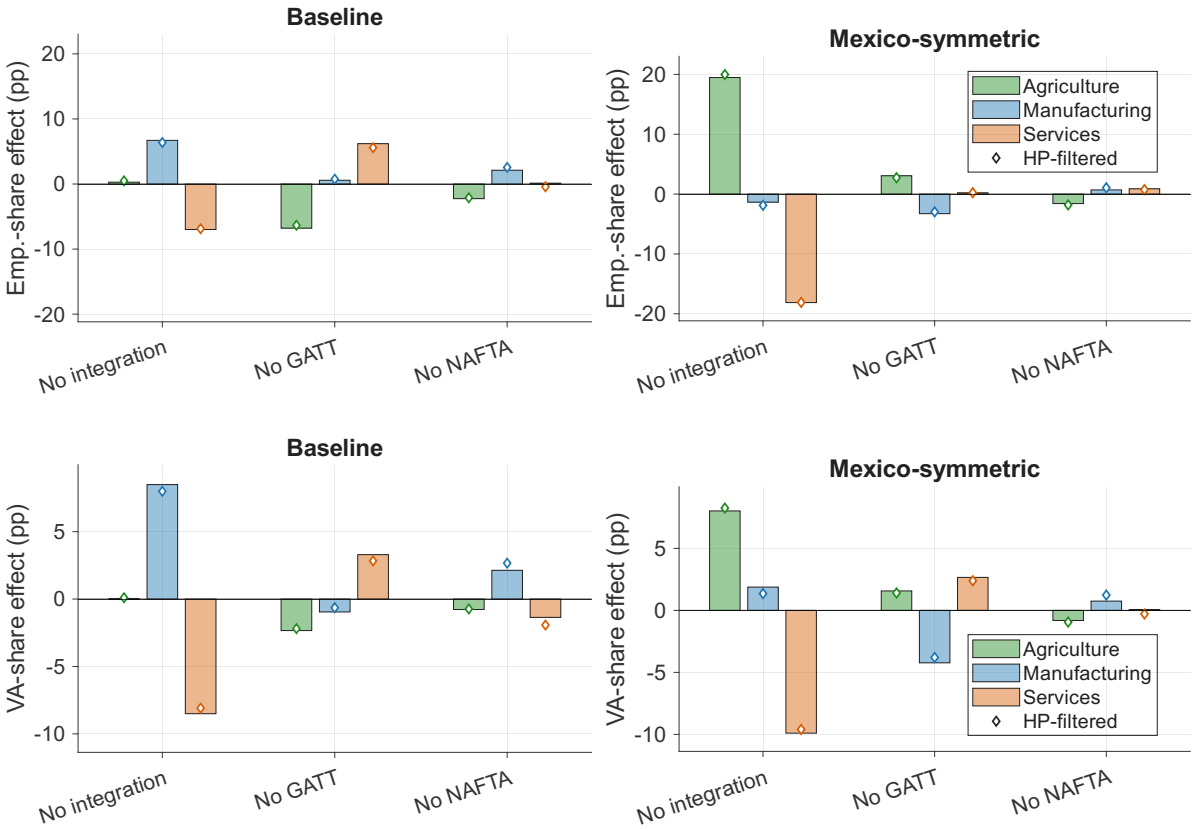


Figure 9: Trade-Cost Effects for Actual and Canada-Level Mexico

Notes: The top row reports the effects of the baseline trade-cost path on 2018 employment shares, measured in percentage points relative to three counterfactuals with frozen trade-cost paths: *no-integration*, *no-GATT*, and *no-NAFTA*. Results are reported for the baseline (left) and *Mexico-symmetric* (right) environments. Bars report results based on the annually calibrated primitives, while diamond markers report equilibrium outcomes recomputed using HP-filtered primitives. Positive values indicate that the observed trade-cost changes increase the sectoral employment share relative to the corresponding frozen-path counterfactual. The bottom row reports the analogous effects on nominal value-added shares.

Table 3: Mexico: Trade-Cost Effects by Partner Development

		Emp.-share effect (pp)			VA-share effect (pp)		
Shock	Year	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
Baseline							
No integration	2010	0.48	4.16	-4.64	0.06	5.76	-5.82
No integration	2018	0.28	6.71	-6.99	0.02	8.49	-8.51
No GATT	2010	-4.51	-1.28	5.79	-1.32	-2.90	4.22
No GATT	2018	-6.78	0.58	6.20	-2.34	-0.96	3.29
No NAFTA	2010	-3.41	1.39	2.02	-1.04	1.06	-0.02
No NAFTA	2018	-2.25	2.12	0.13	-0.77	2.13	-1.36
Mexico-symmetric							
No integration	2010	15.76	-1.59	-14.17	5.74	1.05	-6.79
No integration	2018	19.47	-1.35	-18.13	8.02	1.88	-9.90
No GATT	2010	-0.15	-2.34	2.49	0.07	-3.88	3.81
No GATT	2018	3.06	-3.27	0.21	1.57	-4.23	2.66
No NAFTA	2010	-4.82	1.49	3.32	-2.21	1.38	0.84
No NAFTA	2018	-1.58	0.70	0.88	-0.81	0.75	0.07

Notes: Each effect is the active observed trade-cost path minus the corresponding frozen-path counterfactual. Positive values mean that the trade-cost change raises the indicated sector share. *Mexico-symmetric* assigns Mexico-level initial productivity and other development characteristics to the United States and Canada, while preserving their productivity-growth paths. Results use annual calibrated primitives.

Two features of the observed 1994–2018 window are worth flagging when interpreting these magnitudes. First, NAFTA's tariff schedule was phased in gradually, with most tariff lines eliminated over five to fifteen years rather than immediately in 1994; because the *no-NAFTA* counterfactual freezes trade costs at their 1993 level all the way through 2018, the reported NAFTA effect captures the full post-1994 cumulative trade-cost decline, which also reflects forces unrelated to the agreement itself, including the expansion of global value chains following China's 2001 WTO accession and the 2008–2009 global financial crisis. Second, NAFTA's implementation date coincides closely with the December 1994 Mexican peso crisis, a large macroeconomic shock to Mexican wages, prices, and trade flows in exactly the years used to identify the NAFTA effect. Because sectoral wage wedges, productivity, and trade costs are all recovered from the data year by year, the NAFTA magnitudes reported here should accordingly be interpreted as the total change associated with the observed post-1993 trade-cost path, rather than as an estimate that isolates the treaty's provisions net of its own phase-in schedule or the concurrent macroeconomic crisis.

Figure A.7 traces the estimated effects over the post-1993 period, reflecting the response to the recovered trade-cost path within the model. The effects are particularly pronounced between 1994 and 2000. Although they decline substantially around 2000, they remain economically significant thereafter, broadly following the earlier pattern before gradually diminishing toward 2015–2018.[11] This persistence indicates that changes in trade costs continue to shape sectoral reallocation beyond the initial years of NAFTA.

The development bundle summarize observable differences associated with development, but

[11] Mexico experienced a major reforms to energy, telecommunications, tax policy, education, and labor markets. The energy reform opened the energy sector to private investment and aimed to promote competition and capital reallocation. At the same time, world oil prices collapsed in late 2014, reducing oil-related revenues amid already declining production. These events provide a plausible source of gradually reduction of NAFTA effect after 2014.

each component captures a different aspect of this disparity in levels of development. Technology replacement changes productive efficiency; wage wedges replacement change the relationship between sectoral labor costs; and preference weights replacement change the country-specific demand system. Separating these groups is therefore necessary to interpret the economic content of the joint experiment. We then implement the decomposition analysis to assess the effect of each component.

5.2.3 Decomposition of the Partner Interaction

I use the Shapley (1953) decomposition to assess the effect of each component. This procedure accommodates interactions without assigning them entirely to the group changed last. Let $N = \{T, \xi, \omega\}$ denote the three groups of characteristics, and let $S \subseteq N$ indicate the groups replaced with their Mexican counterparts. The empty set gives baseline, while $S = N$ gives the *Mexico-symmetric*. I solve all eight environments under both trade-cost regimes. For a given sector and outcome, let $I(S)$ the effect of observed variation in trade costs within a given partner environment S as defined in Equation (25). The contribution of group $g \in N$ to the difference between these integration effects across the baseline and *Mexico-symmetric* environments, $I(\varnothing) - I(N)$ is calculated as:[12]

$$\mathcal{C}_g = \sum_{S \subseteq N \setminus \{g\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left[I(S) - I(S \cup \{g\}) \right]. \tag{27}$$

I adopt the actual-minus-full-replacement convention throughout: a positive contribution means that retaining the group's actual partner characteristics raises the integration response relative to replacing them, on average across replacement orders. A negative contribution means the opposite.

Panel A of Table 4 reports the 2018 integration effects within the eight partner environments, including the baseline and *Mexico-symmetric* environments. Panel B allocates the difference between the baseline and full-replacement integration effects using the Shapley decomposition. The two panels thus report within-environment effects and contributions to a between-environment gap, respectively.

The factorial comparisons indicate that differences in sectoral technology account for most of the role of partner development in shaping the effects of trade integration on Mexico's structural change documented in Section 5.2.2. Replacing partners' sectoral technology alone generates responses close to those obtained under the full replacement of development characteristics. The Shapley decomposition in Panel B of Table 4 confirms the dominant contribution of technology, while indicating smaller contributions from sectoral wage wedges and preference weights.

The manufacturing sector provides a particularly informative illustration of this mechanism, given its central role in the literature on trade and structural change. Replacing partners' sectoral technology alone reverses the effect of trade integration on Mexico's manufacturing employment share, from +6.71 to −0.38 percentage points. Replacing technology and wage wedges jointly, while retaining partners' original preference weights, produces a similar effect of −0.45 points. By contrast, replacing wage wedges or preferences alone yields positive effects of 7.85 and 6.64 points, respectively. These comparisons indicate that differences in sectoral technology are sufficient to reverse the manufacturing employment response, without requiring changes in partners' preference weights.

The Shapley decomposition quantifies the relative contributions of these characteristics while accounting for their interactions. Of the 8.06-percentage-point difference between the baseline

[12] By construction, $\sum_g \mathcal{C}_g = I(\varnothing) - I(N)$. The decomposition is an accounting exercise within the calibrated model. I report the eight underlying integration effects as well as the decomposition. This makes clear whether a particular result is present when a group is changed alone or arises through its interaction with other characteristics.

Table 4: Integration Effects and Contributions of Each Component, 2018

	Emp. share			VA share		
	Agr.	Mfg.	Serv.	Agr.	Mfg.	Serv.
Panel A. Integration effects within each partner environment						
Actual partners (*Baseline*)	0.28	6.71	-6.99	0.02	8.49	-8.51
Technology only	16.73	-0.38	-16.35	6.67	2.84	-9.51
Wedges only	-4.58	7.85	-3.26	-1.45	8.59	-7.14
Technology and wedges	16.68	-0.45	-16.24	6.61	2.64	-9.25
Preferences only	0.43	6.64	-7.07	0.06	8.44	-8.51
Technology and preferences	19.57	-1.33	-18.24	8.11	2.03	-10.14
Wedges and preferences	-4.54	7.81	-3.27	-1.44	8.56	-7.12
Full replacement (*Mexico-symmetric*)	19.47	-1.35	-18.13	8.02	1.88	-9.90
Panel B. Contributions of each component						
Technology	-20.22	8.13	12.10	-8.06	6.17	1.89
Wedges	2.49	-0.56	-1.93	0.78	0.03	-0.81
Preferences	-1.46	0.49	0.97	-0.73	0.41	0.32
Baseline minus *Mexico-symmetric*	-19.19	8.06	11.13	-8.01	6.61	1.40

Notes: All entries are in percentage points. Panel A reports observed-cost minus frozen-cost sectoral shares within each partner environment; the freeze holds Mexico's bilateral trade costs at 1971 levels. Replacements apply to the United States and Canada, and technology replacements preserve each partner's own sectoral growth. Panel B reports Shapley contributions to the actual-partner effect minus the full-replacement effect in Panel A. Its total is this difference, not the full-replacement row. Contributions average over all replacement orders and add up before rounding. A positive contribution increases the actual-minus-replacement gap; it need not have the same sign as the corresponding integration effect in Panel A.

and full-replacement manufacturing employment effects, sectoral technology contributes 8.13 points, wage wedges contribute -0.56 points, and preferences contribute 0.49 points. The corresponding contributions to manufacturing value-added shares are 6.17, 0.03, and 0.41 points, respectively. The technology contribution exceeds the net employment gap because the wage-wedge contribution partially offsets it. Importantly, the relatively small contribution of preference replacement does not imply that income effects are negligible, since nonhomothetic demand operates in every counterfactual, including those with unchanged preference weights. Similarly, wage-wedge contributions capture differences in sectoral labor costs and the relationship between employment and value added, rather than the causal effects of removing labor-market distortions. Taken together, these results identify sectoral technology differences as the principal channel through which partner development shapes Mexico's manufacturing response to trade integration.

The sectoral technology differences may affect structural change through different mechanisms. Differences in overall productivity influence production costs and equilibrium purchasing power, whereas differences in relative sectoral productivity alter patterns of comparative advantage and specialization. To distinguish these mechanisms, I decompose partner technology into a common-level component and a relative sectoral component. I then quantify how each component shapes the effects of trade integration on Mexico's sectoral employment and value-added shares.

5.2.4 Technology Levels and Relative Sectoral Productivity

To distinguish the level of productive efficiency from its sectoral pattern, define the initial technology replacement factor for partner j as

$$r_{jk} = \frac{T_{\text{MEX},k,1971}}{T_{jk,1971}}, \qquad \bar{r}_j = \exp\left(\frac{1}{3}\sum_k \log r_{jk}\right), \qquad \widetilde{r}_{jk} = \frac{r_{jk}}{\bar{r}_j}. \tag{28}$$

A common-level replacement multiplies all of partner j's sectoral technologies by $\bar{r}_j$, preserving the relative technology profile. A profile replacement multiplies sector k by $\widetilde{r}_{jk}$, preserving the equal-weight geometric mean of the technology scales. Applying both factors yields the full technology replacement.

The common technology-level replacement scales each partner's technology proportionally across all three sectors, preserving its relative sectoral productivity profile. This experiment examines how trade integration would affect Mexico's sectoral allocation if its partners had a lower overall technology level but retained their original relative productivity differences across sectors. By contrast, the relative productivity-profile replacement changes the distribution of technology across sectors while preserving each partner's equal-weight geometric mean of sectoral technology levels. This experiment examines how trade integration would affect Mexico's sectoral allocation if its partners retained their overall technology level but adopted Mexico's initial relative sectoral productivity profile. By altering relative productivity across sectors, this experiment isolates the role of comparative advantage in shaping trade patterns and sectoral reallocation.[13]

Together, these experiments distinguish the contribution of differences in overall technology levels from that of relative sectoral productivity differences, allowing me to assess whether these two dimensions reinforce or offset each other in shaping Mexico's structural response to trade integration.

Finally, I consider intermediate partner configurations indexed by $\alpha \in [0,1]$. For the full bundle, the positive primitives are interpolated in logs:

$$\begin{aligned} T_{jk,t}(\alpha) &= T_{jk,t} r_{jk}^{\alpha}, \\ \xi_{jk,t}(\alpha) &= \xi_{jk,t}\left(\frac{\xi_{\text{MEX},k,t}}{\xi_{jk,t}}\right)^{\alpha}, \\ \omega_{jk}(\alpha) &= \omega_{jk}\left(\frac{\omega_{\text{MEX},k}}{\omega_{jk}}\right)^{\alpha}. \end{aligned} \tag{29}$$

Thus $\alpha = 0$ gives baseline and $\alpha = 1$ gives *Mexico-symmetric*. I also trace technology-only, common-level, and profile replacements while leaving the remaining characteristics unchanged. No additional normalization of preference weights is imposed along the path. The curves describe sensitivity along specified configurations of primitives.[14]

Table 5 assesses whether the role of trading partners' technology in shaping the effects of trade integration is driven primarily by differences in overall productivity levels or by relative sectoral productivity profiles. The results indicate that replacing the common technology level produces integration effects relatively close to those obtained under the baseline, whereas replacing the relative sectoral productivity profile generates substantially larger changes, comparable to those obtained under the full technology replacement. This finding suggests that differences in comparative advantage, as captured by relative sectoral productivity profiles, are the principal channel through which partner development shapes Mexico's structural response to trade integration.

[13] These experiments use an equal-weight geometric-mean normalization. They do not hold aggregate GDP or an expenditure-weighted productivity index constant, as these outcomes adjust endogenously in equilibrium.

[14] A change in sign along one curve establishes a conditional numerical result, rather than a universal income threshold.

The manufacturing employment response illustrates this mechanism. Replacing only the common technology level reduces the effect of trade integration from 6.71 percentage points under the baseline to 5.09 points, leaving it positive. By contrast, replacing only the relative sectoral productivity profile reverses the effect to -3.60 percentage points. Replacing both components yields an effect of -0.38 points. These results indicate that differences in relative sectoral productivity, rather than overall productivity levels, are primarily responsible for the reversal in Mexico's manufacturing employment response.

The sign reversal is therefore associated with the relative sectoral profile in these experiments, rather than with a common reduction in partner technology alone. The two components do not contribute additively: changing the common level also changes wages, expenditure, and the response to the profile replacement. The profile-only experiment leaves the equal-weight geometric mean of partner technology unchanged, but does not hold partner GDP or real consumption fixed.

The underlying replacement factors clarify this distinction. In productivity-draw units, the U.S. technology replacement multiplies initial agricultural, manufacturing, and services efficiency by 0.108, 0.670, and 0.455, respectively; the Canadian factors are 0.119, 0.722, and 0.547. The corresponding common-level factors are 0.320 and 0.361. Dividing the sectoral factors by these common components therefore reduces relative agricultural efficiency and raises relative manufacturing efficiency in the two partners. This changes Mexico's comparative position across sectors. The full replacement is a large counterfactual, especially in agriculture, which motivates examining intermediate configurations as well as the endpoints.

Table 5: The Level and Sectoral Composition of Partner Technology

	Employment			Value added		
	Agr.	Mfg.	Serv.	Agr.	Mfg.	Serv.
Actual partners (*Baseline*)	0.28	6.71	-6.99	0.02	8.49	-8.51
Full technology replacement	16.73	-0.38	-16.35	6.67	2.84	-9.51
Uniform productivity scale	1.81	5.09	-6.89	0.54	6.85	-7.39
Relative productivity profile	26.47	-3.60	-22.87	13.88	-3.09	-10.79
Technology and wedges	16.68	-0.45	-16.24	6.61	2.64	-9.25

Notes: Integration effects in percentage points. The common-level and relative-profile replacements are defined in Equation (28). Both retain the partners' own wage wedges and preference weights. The final row also replaces wage wedges while retaining preference weights. The separate level and profile effects need not add to the full technology effect.

The nominal value-added comparisons reinforce the distinction. Its manufacturing integration effect remains positive under the common-level replacement, at 6.85 percentage points, but becomes negative under the profile-only replacement, at -3.09 points. The agricultural employment response is 1.81 points in the former experiment and 26.47 points in the latter. Thus a summary measure of partner productive efficiency conceals differences in comparative advantage that are quantitatively consequential for sectoral allocation. This result complements the interaction of trade and sector-biased productivity emphasized by Sposi et al. (2026) and the transmission of foreign sectoral productivity studied by De Vries et al. (2026). The present exercise isolates the partner profile while holding the trade-cost experiment fixed; it does not endogenize technology adoption as in Fujiwara & Matsuyama (2024).

Figure 10 traces the responses at eleven equally spaced replacement intensities. Along the full-bundle path, the manufacturing employment effect changes from 0.36 percentage points at $\alpha = 0.6$ to -0.36 points at $\alpha = 0.7$. The corresponding sign change occurs between 0.5 and 0.6 along the technology-only path and between 0.4 and 0.5 along the profile-only path. The common-level path remains positive at all evaluated points. These comparisons show that the endpoint contrast is part of a broader response pattern within the calibrated model.

The response is nonlinear and need not be monotonic. In particular, the technology-only and profile-only manufacturing employment effects become slightly less negative near the endpoint after reaching their minima at intermediate configurations. The sign-change intervals are properties of these specified paths. They do not identify an income threshold that applies across countries, alternative normalizations, or other trade-cost changes.

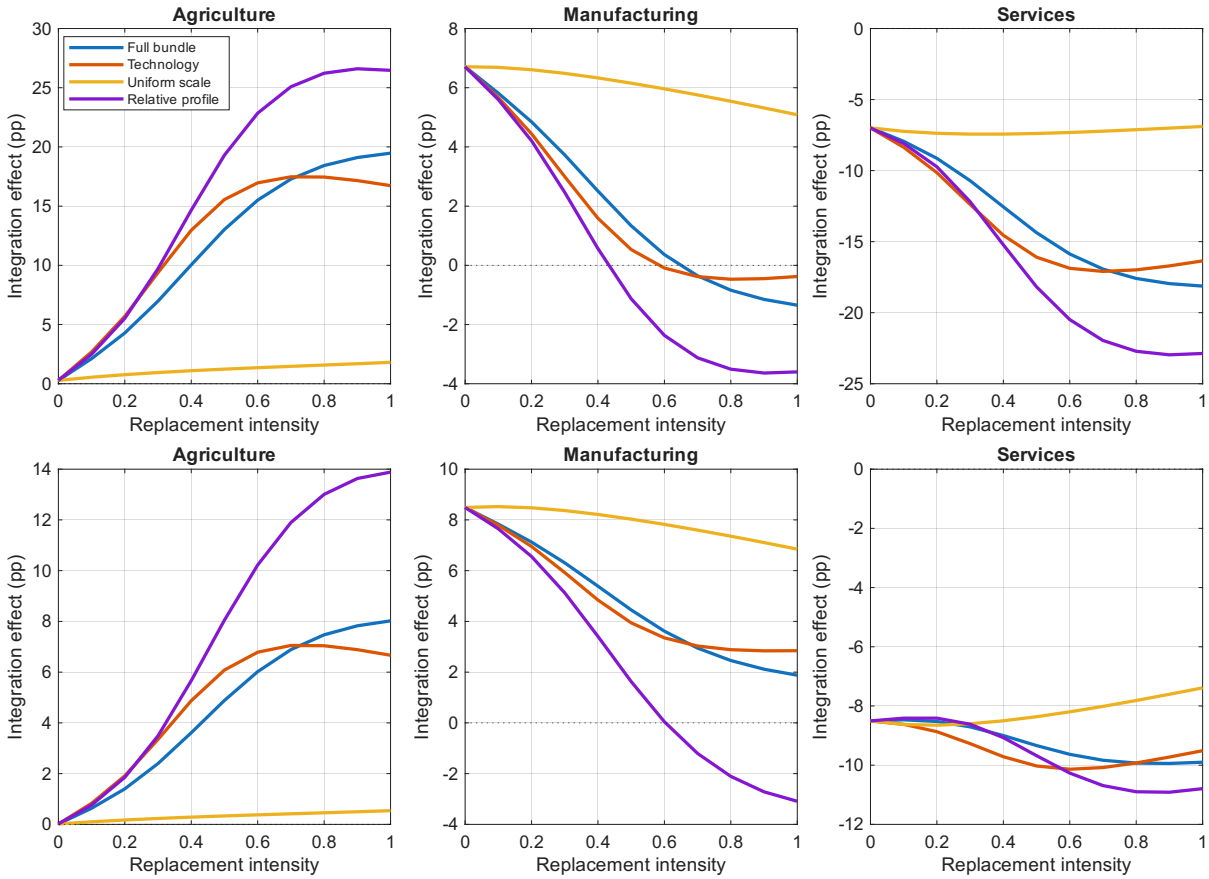


Figure 10: Integration Effects along Alternative Partner Configurations

Notes: Employment effects in the top row and nominal value-added effects in the bottom row, in percentage points. At $\alpha = 0$, all paths have actual partner characteristics. At $\alpha = 1$, each implements its specified replacement. Lines connect eleven computed equilibrium pairs; they are not estimated response functions or confidence intervals. Technology-only, common-level, and profile-only paths retain partner wage wedges and preference weights.

The important finding is that changing the common technology level leaves the manufacturing response positive, whereas changing the sectoral profile reverses its sign. The two components interact, so their effects should not be added together. Economically, the actual U.S. and Canadian productivity advantages over Mexico are particularly pronounced in agriculture. Replacing their relative productivity profiles changes Mexico's comparative position across sectors. The resulting adjustment in trade, prices, and production can redirect Mexico's integration-induced employment response toward agriculture and away from manufacturing.

Taken together, these findings indicate that the effects of trade integration depend on both trading partners' overall productive efficiency and their relative sectoral productivity profiles. In the Mexican case, the latter plays a particularly important role in determining how integration reallocates employment and value added across sectors. Partner development therefore shapes structural change primarily through differences in comparative advantage, rather than through aggregate productivity or income gaps alone.

The preceding experiments hold trading partners' labor endowments fixed, thereby isolating

the role of their development characteristics from that of market size. As a final exercise, I impose symmetric labor endowments across countries to assess whether differences in market size alter the effects of partner development on Mexico's structural response to trade integration.

5.2.5 Partner Size versus Partner Development

The symmetric-partner experiments above hold U.S. and Canadian labor endowments fixed at their observed levels while replacing their development bundles. A natural complementary question is how much of Mexico's structural change response to trade policy reflects the size of its trading partners, rather than differences in their development characteristics, such as productivity, wedge wages, and expenditure patterns. To isolate this channel, I conduct an additional experiment in which I replace only the U.S. and Canadian labor endowments with Mexico's observed labor-endowment path, while holding their productivity, wage wedges, preference weights, transfer-rule coefficients, and bilateral trade costs at their baseline values. Country wage levels continue to be solved endogenously. I then repeat trade cost shock experiments in this symmetric population size environment.

Table 6 reports the effects of changing partner labor endowments. The full integration experiment changes baseline employment shares by $(+0.28, +6.71, -6.99)$ percentage points in agriculture, manufacturing, and services. With U.S. and Canadian labor endowments set to Mexico's path, the effects are $(+0.38, +5.45, -5.83)$ points. The manufacturing effect in the post-1993 North American window is +1.30 points, compared with +2.12 points in the baseline; the post-1985 effect is +0.22 points, compared with +0.58 points. Partner labor endowments therefore slightly affect the magnitude of the responses.

Table 6: Mexico: 2018 Trade Effects by Development and Partner Size

		Emp.-share effect (pp)			VA-share effect (pp)		
Shock	Environment	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
Mexico integration	Baseline	0.28	6.71	-6.99	0.02	8.49	-8.51
Mexico integration	Mexico-symmetric	19.47	-1.35	-18.13	8.02	1.88	-9.90
Mexico integration	U.S./Canada pop. = Mexico	0.38	5.45	-5.83	0.06	6.94	-7.00
GATT	Baseline	-6.78	0.58	6.20	-2.34	-0.96	3.29
GATT	Mexico-symmetric	3.06	-3.27	0.21	1.57	-4.23	2.66
GATT	U.S./Canada pop. = Mexico	-5.70	0.22	5.48	-1.95	-1.10	3.06
NAFTA	Baseline	-2.25	2.12	0.13	-0.77	2.13	-1.36
NAFTA	Mexico-symmetric	-1.58	0.70	0.88	-0.81	0.75	0.07
NAFTA	U.S./Canada pop. = Mexico	-1.62	1.30	0.31	-0.55	1.26	-0.71

Notes: Each effect is the active observed trade-cost path minus the corresponding frozen-path counterfactual in 2018; positive values mean that the observed trade-cost change raises the sector share. The table compares the baseline, *Mexico-symmetric*, and population-symmetric environments. The population-symmetric environment replaces only U.S. and Canadian labor endowments with Mexico's observed path, holding their development characteristics, productivity-growth paths, and trade-cost paths fixed. Results use annual calibrated primitives.

Taken together, the counterfactual results show that Mexico's structural response to trade integration cannot be summarized by the observed decline in trade costs alone. The same trade-cost paths generate different sectoral reallocations when the development characteristics of Mexico's partners are changed, and the effects are largest when the counterfactual changes Mexico's exposure to the advanced U.S. market. The partner-size experiment shows that development and market size are distinct but interacting margins. These results motivate the discussion below, which interprets the economic mechanism and situates the findings within the broader literature on trade and structural change.

6 Discussion

The counterfactual experiments establish that the effects of trade integration on structural change depend on the development characteristics of trading partners. In the Mexican case, replacing the United States and Canada with partners sharing Mexico's development characteristics generates substantially different responses in sectoral employment and nominal value-added shares, even when bilateral trade-cost paths, partner labor endowments, input-output linkages, trade imbalances, and sector-specific productivity-growth paths are held fixed. Thus, identical reductions in trade costs need not produce the same pattern of structural change. The decomposition further reveals that differences in partners' sectoral technology account for most of this heterogeneity, while sectoral wage wedges and preference weights play smaller roles.

An important distinction is between the effects of partner characteristics on sectoral allocations and their effects on the response to trade integration. Differences in employment or value-added shares between the baseline and counterfactual partner configurations do not, by themselves, identify the role of partner development in shaping integration effects, since these differences may also arise in the absence of further trade-cost reductions. The interaction defined in Equation (26) addresses this distinction by comparing the effects of the same trade-cost changes across alternative partner configurations. It therefore isolates how partner development conditions the structural response to integration, rather than simply measuring differences in equilibrium allocations.

The technology decomposition provides further insight into the underlying mechanism. Trading partners differ not only in their overall productive efficiency but also in the distribution of productivity advantages across sectors. While differences in overall productivity influence production costs, incomes, and expenditure, differences in relative sectoral productivity shape comparative advantage and patterns of specialization. The counterfactual results indicate that the latter dimension is particularly important for Mexico. Replacing partners' relative sectoral productivity profiles substantially changes the manufacturing employment response to integration, whereas replacing their common technology levels produces a response closer to the baseline. Consequently, the role of partner development cannot be adequately summarized by differences in aggregate productivity or income alone. It depends critically on the sectoral structure of comparative advantage.

The partner-size experiment highlights an additional dimension of this relationship. Development characteristics and market size influence trade integration through distinct but interacting channels. While sectoral productivity and expenditure patterns shape relative costs and the composition of foreign demand, partner labor endowments influence the scale of foreign markets. Imposing symmetric labor endowments changes the magnitude and, in some cases, the sectoral pattern of Mexico's response to trade-cost reductions. This finding underscores the importance of distinguishing the characteristics of trading partners from their market size when evaluating the structural consequences of integration.

These findings contribute to the quantitative literature on trade and structural change by showing that the sectoral effects of trade-cost reductions depend on the characteristics of the economies with which a country integrates. Mexico's integration with the United States and Canada illustrates how access to advanced economies interacts with domestic comparative advantage, intermediate-input costs, nonhomothetic demand, and uneven sectoral productivity growth to shape the allocation of economic activity. Trade integration therefore cannot be characterized solely by the magnitude of trade-cost reductions; its sectoral consequences also depend on the productivity profiles and economic characteristics of trading partners.

This interpretation complements the findings of Daymard (2025), who emphasizes the importance of domestic agricultural productivity and finds that structural transformation in most countries remains broadly consistent with closed-economy mechanisms. The Mexican experience identifies an additional channel through which international trade can influence structural

change, particularly in economies closely integrated with more developed trading partners. It does not imply that partner characteristics dominate domestic drivers of structural transformation across countries. Rather, it establishes that the development characteristics of trading partners are an important determinant of how trade integration affects sectoral reallocation.

7 Conclusion

This paper examines how the characteristics of trading partners affect the sectoral consequences of integration in Mexico. Using a multi-country and multi-sector model with nonhomothetic preferences, sectoral productivity differences and international production linkages, I compare the response to the same trade-cost path across alternative configurations of North American partners. The analysis separates differences in sectoral allocations from differences in integration effects and decomposes the latter into technology, wage-wedge, and preference contributions.

The counterfactual results reveal that trading partners' development characteristics substantially shape the sectoral effects of trade integration. Under the baseline configuration, integration increases Mexico's manufacturing value-added share by 8.49 percentage points in 2018, compared with only 1.88 points when the United States and Canada are assigned Mexico's development characteristics. Conversely, the agricultural value-added response increases from 0.02 to 8.02 percentage points. The differences are even more pronounced for employment: integration raises manufacturing's employment share by 6.71 percentage points and agriculture's by only 0.28 points under the baseline, whereas the corresponding effects under the *Mexico-symmetric* counterfactual are -1.35 and $+19.47$ points. These findings demonstrate that integration with more developed partners favors manufacturing expansion in Mexico, while integration with similarly developed partners would instead induce a substantial reallocation toward agriculture. Technology accounts for the principal difference. Separating technology into a common level and a relative sectoral profile shows that the latter is central to the manufacturing employment reversal.

The broader implication is that the development gap between trading partners is an incomplete description of their influence on structural change. The level and sectoral composition of foreign productivity, together with expenditure and labor-cost characteristics, determine how integration changes domestic sourcing, producer revenues, and wages. Distinguishing these objects provides a more precise account of the conditions under which trade integration changes the allocation of labor across agriculture, manufacturing, and services.

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A Supplement Results

A.1 Calibrated productivity growth

Table A.1 reports calibrated productivity growth for all countries in the model, and Figure A.1 plots them. I report the implied Frechet productivity index $A_{ik,t} = \gamma(1-1/\theta)T_{ik,t}^{1/\theta}$, rather than the Frechet location parameter $T_{ik,t}$ itself. The reason is that $T_{ik,t}$ governs the distribution of variety-level productivity draws, while $A_{ik,t}$ expresses the corresponding average efficiency scale in productivity units. Since the table normalizes each country-sector cell by its 1971 value, the constant $\gamma(1-1/\theta)$ cancels and the reported index satisfies $A_{ik,t}/A_{ik,1971} = (T_{ik,t}/T_{ik,1971})^{1/\theta}$. Productivity growth is sector-biased in every country. Agriculture displays the largest cumulative increase, while manufacturing grows more moderately. Mexico's manufacturing productivity index reaches 1.84 by 2018, compared with 2.51 in Canada and 2.68 in the United States. Services productivity in Mexico rises by a factor of 2.89, below the United States but close to Canada.

Table A.1: Calibrated Productivity Growth Across Countries

	Index, 2018 (1971=1)			Average annual log growth (%)		
Country	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
Mexico	4.51	1.84	2.89	3.21	1.29	2.26
Canada	3.25	2.51	3.15	2.51	1.96	2.44
United States	3.49	2.68	3.49	2.66	2.10	2.66
Rest of World	3.85	2.33	2.48	2.87	1.80	1.93

Notes: The table reports the implied Frechet productivity index $A_{ik,t}$, where $A_{ik,t} = \Gamma(1-1/\theta)T_{ik,t}^{1/\theta}$. The index columns report $A_{ik,2018}/A_{ik,1971}$. Average annual log growth is computed over 1971–2018.

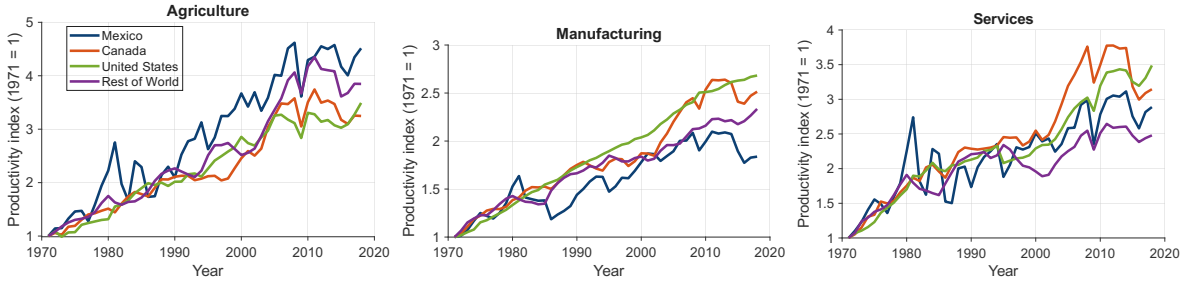


Figure A.1: Development Bundle: Sectoral Productivity Paths

Notes: The panels plot the Frechet productivity index $A_{ik,t}$, normalized to 1971 within each country-sector cell. Lines compare Mexico, Canada, the United States, and the rest of the world.

A.2 Development-Bundle Diagnostics

Figures A.2 and A.3 report the country-level series that enter the development-bundle diagnostics. These diagnostics show that the bundle is not an arbitrary relabeling of countries. Canada and the United States differ from Mexico in productivity levels, productivity growth, and sectoral wage wedges. The counterfactual replacement uses these diagnostics asymmetrically: it replaces productivity levels but preserves the target country's productivity-growth path, while wage wedges and preference weights are reassigned as development characteristics. Country manufacturing wages are shown only as calibration diagnostics and are not reassigned as exogenous paths.

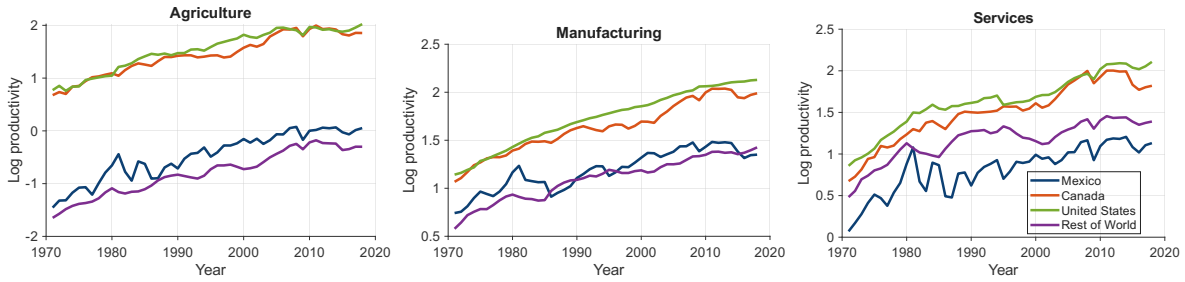


Figure A.2: Development Bundle: Log Productivity Levels

Notes: The panels plot $\log(A_{ik,t})$ separately for agriculture, manufacturing, and services. Each panel contains one line for Mexico, Canada, the United States, and the rest of the world. These paths are not normalized to 1971 and therefore retain cross-country level differences under the model's productivity normalization.

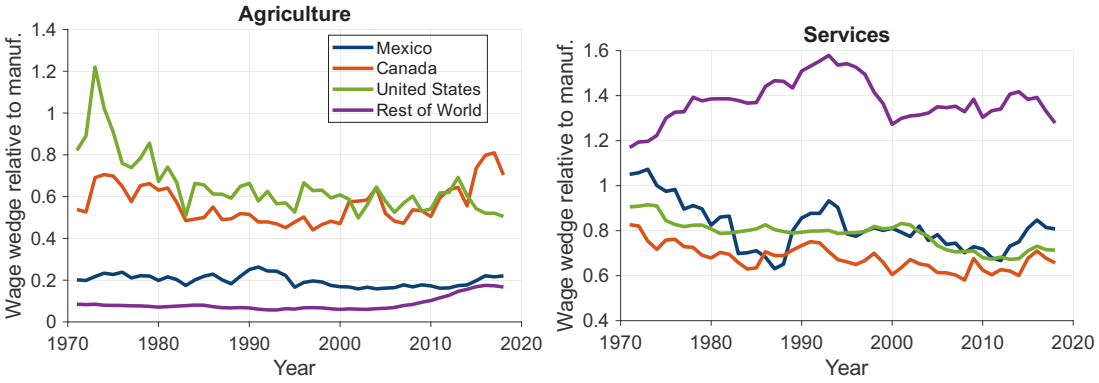


Figure A.3: Development Diagnostics: Sectoral Wage Wedges

Notes: The first two panels plot agriculture and services wage wedges relative to manufacturing. The manufacturing wedge is normalized to one by construction. Lines compare Mexico, Canada, the United States, and the rest of the world.

A.3 Supplemental Model Fit

Figure A.4 and Table A.2 compare observed and model-implied nominal gross-output shares. Model gross output is measured by domestic supplier revenue, including exports. Across country-sector cells, the RMSE is 2.94 percentage points and the mean correlation is 0.855. This assesses production and input–output consistency, but is not independent validation: gross output enters the recovered price indexes, input shares, and domestic absorption data.

Table A.2: Supplemental Fit: Nominal Gross-Output Shares

	RMSE (pp)			Correlation		
Country	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
Mexico	1.08	4.46	3.92	0.958	0.223	0.874
Canada	0.33	2.51	2.70	0.933	0.880	0.911
United States	0.30	4.30	4.56	0.977	0.994	0.994
Rest of World	0.60	2.48	2.81	0.945	0.743	0.821
All countries		2.94			0.855	

Notes: RMSE is the root mean squared error in percentage points between model and data shares over 1971–2018. Observed shares are nominal gross output by sector divided by total nominal gross output in the country-year. Model gross output is measured by supplier revenue, $\sum_j P_{jk,t} Q_{jk,t} \pi_{jik,t}$, which is the model counterpart to nominal gross output.

The supplemental fit is strongest for the broad long-run movements in agriculture and services, and remains informative for manufacturing despite larger annual deviations. This pattern is consistent with the model's role in the paper: it disciplines comparative equilibrium mechanisms rather than absorbing every short-run movement in sectoral production. The gross-output diagnostic therefore provides additional support for the intermediate-input and trade-revenue blocks, which generate plausible sectoral production shares without making those shares separate calibration targets.

A.4 Which Advanced Partner Matters for Mexico?

Table A.3: Mexico: Which Advanced Partner Drives the Development Effect?

		Emp.-share effect (pp)			VA-share effect (pp)		
Replacement	Year	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
U.S. and Canada	2010	-18.98	6.84	12.15	-6.81	5.50	1.31
U.S. and Canada	2018	-21.77	9.19	12.58	-8.89	7.63	1.26
U.S. only	2010	-12.25	4.78	7.46	-4.09	3.86	0.23
U.S. only	2018	-14.46	6.75	7.71	-5.49	5.62	-0.13
Canada only	2010	-0.91	0.33	0.58	-0.27	0.22	0.05
Canada only	2018	-0.89	0.41	0.48	-0.30	0.30	-0.00

Notes: Effects are baseline minus the indicated *Mexico-symmetric* environment. Positive values mean that the actual advanced partner raises Mexico's sector share relative to the Mexico-level-partner environment. The joint effect need not equal the sum of the single-partner effects because prices, expenditures, and trade shares adjust in general equilibrium. Results use annual calibrated primitives.

Table A.3 reports separate U.S. and Canadian replacement experiments. Baseline employment shares minus the U.S.-only replacement are $(-14.46, +6.75, +7.71)$ percentage points in

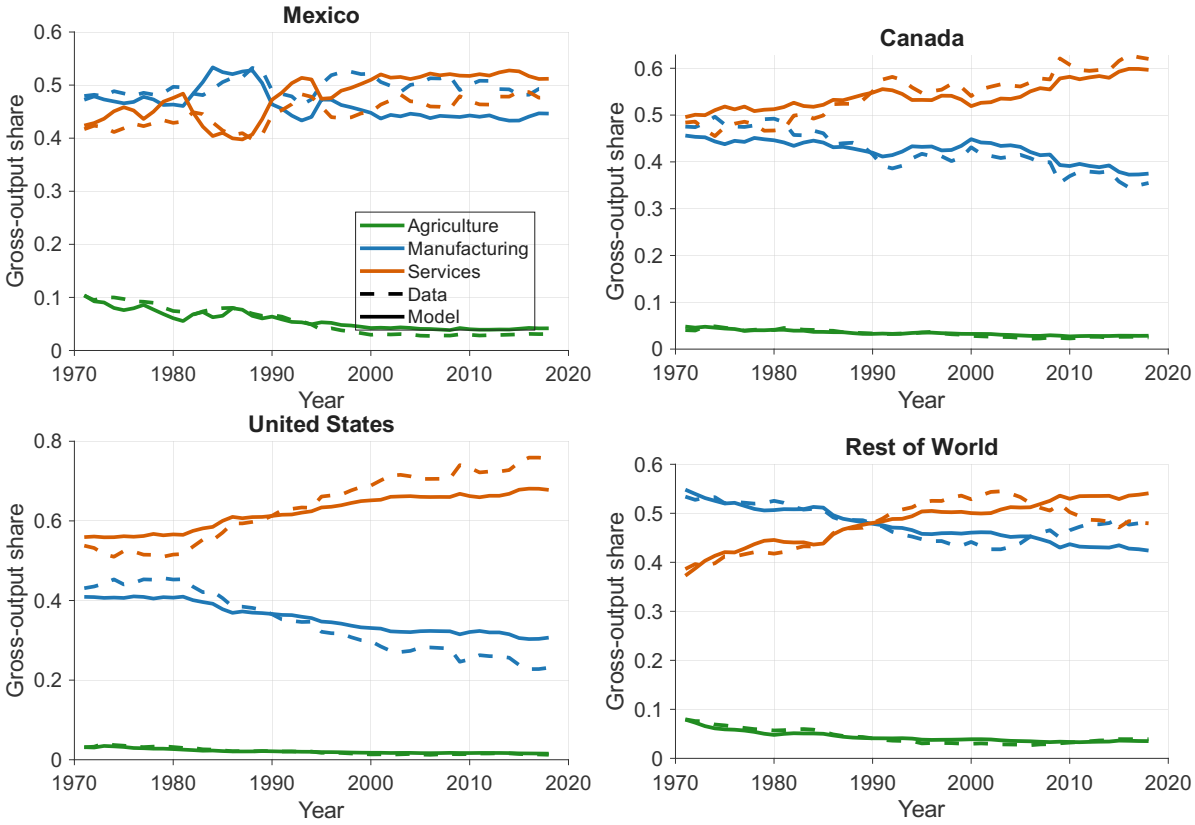


Figure A.4: Supplemental Fit: Sectoral Gross-Output Shares

Notes: The figure plots nominal gross-output shares by sector and country. Dashed lines are observed shares and solid lines are model-implied shares. Model gross output is measured as sectoral supplier revenue, $\sum_j P_{jk,t} Q_{jk,t} \pi_{jik,t}$. Colors identify agriculture, manufacturing, and services.

2018; the Canada-only comparison gives $(-0.89, +0.41, +0.48)$ points. The corresponding nominal value-added differences are $(-5.49, +5.62, -0.13)$ and $(-0.30, +0.30, 0.00)$ points. These are differences in allocation levels under observed trade costs, not an additive decomposition of integration effects. The U.S. replacement has the larger separate effect, and general equilibrium interactions account for the difference between the joint and summed separate comparisons.

A.5 Supplementary results: Trade Policy experiments

This subsection reports the full time paths behind the trade-policy and trade-integration effects summarized in the main text. For each development environment, the figures plot employment-share effects in the top row and value-added-share effects in the bottom row. Lighter lines use the annual calibrated primitives, while darker lines report the same counterfactual exercises after HP filtering the exogenous primitives with $\lambda = 6.25$.

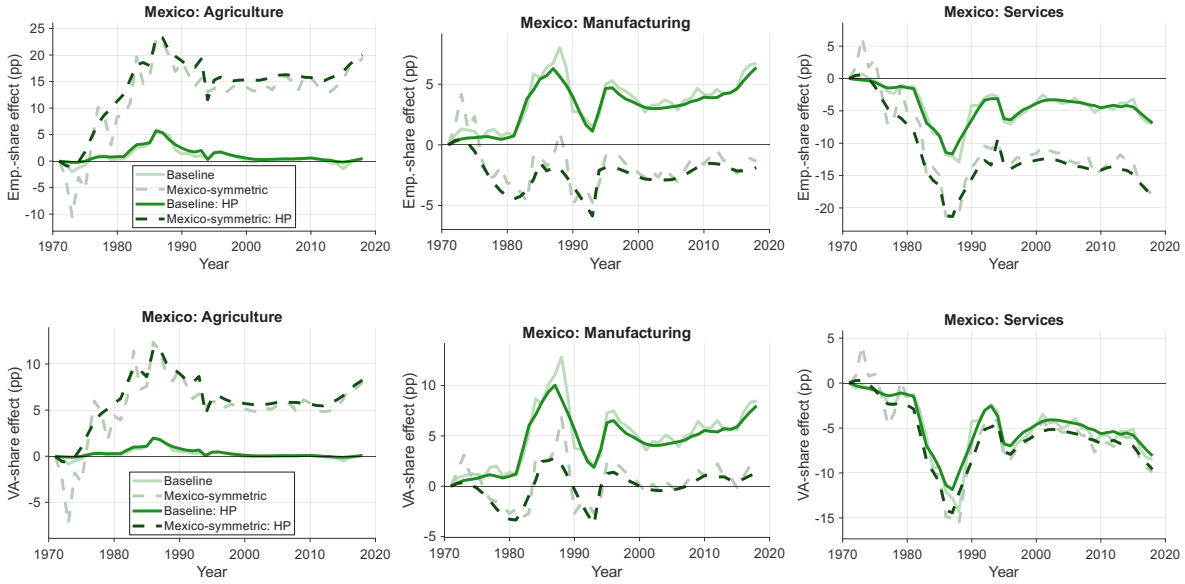


Figure A.5: Mexico-Integration Effects by Development Environment

Notes: The figures plot broad Mexico-integration effects on employment and value-added shares in the baseline and *Mexico-symmetric* environments. The top row shows employment while the bottom row shows value-added shares. Each effect is the active observed trade-cost path minus the no-integration frozen-path counterfactual, which holds bilateral trade costs involving Mexico at their 1971 levels. Solid lines report actual Mexico and dashed lines report the counterfactual development environment; lighter lines use annual calibrated primitives and darker lines use HP-filtered primitive simulations.

A.6 Partner Size and Development Environments

Table A.4 reports the complete 2010 and 2018 entries for the partner-size exercise. The table keeps the same environments as the main-text table: baseline, *Mexico-symmetric* and population-symmetric. This presentation separates two comparisons. The first compares the baseline with *Mexico-symmetric*. The second compares the baseline with the population-symmetric environment, which changes partner size while holding partner development characteristics fixed.

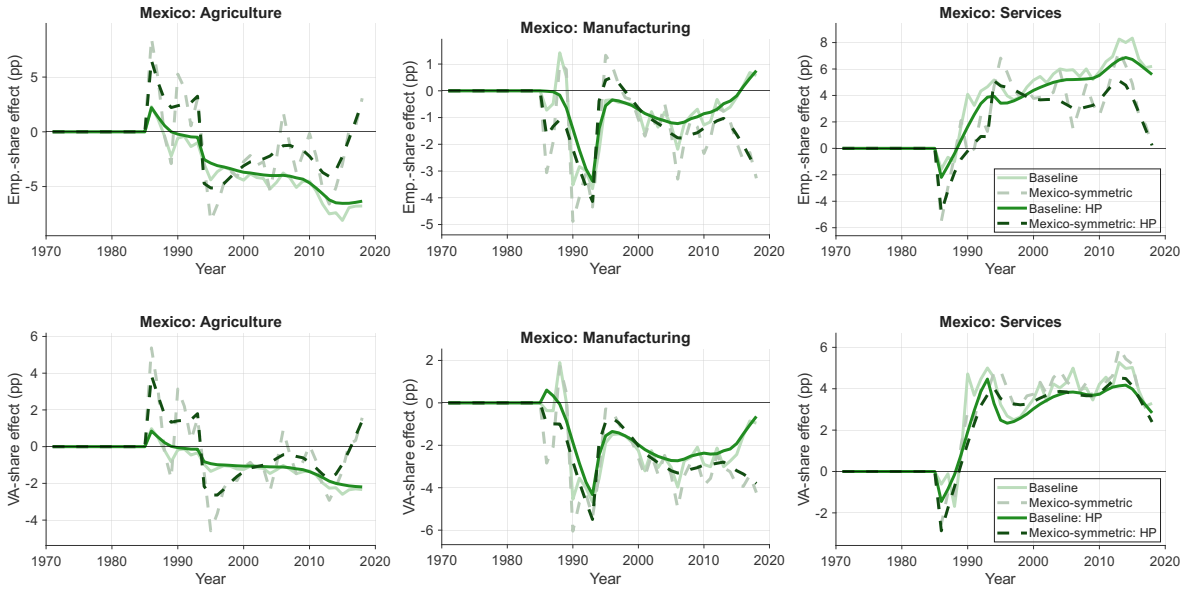


Figure A.6: GATT Effects by Development Environment

Notes: The figures plot GATT effects on employment and value-added shares in the baseline and *Mexico-symmetric*. The top row shows employment, while the bottom row shows value-added shares. Each effect is the active observed post-1986 trade-cost path minus the *no-GATT* frozen-path counterfactual. Solid lines report actual Mexico and dashed lines report the counterfactual development environment; lighter lines use annual calibrated primitives and darker lines use HP-filtered primitive simulations.

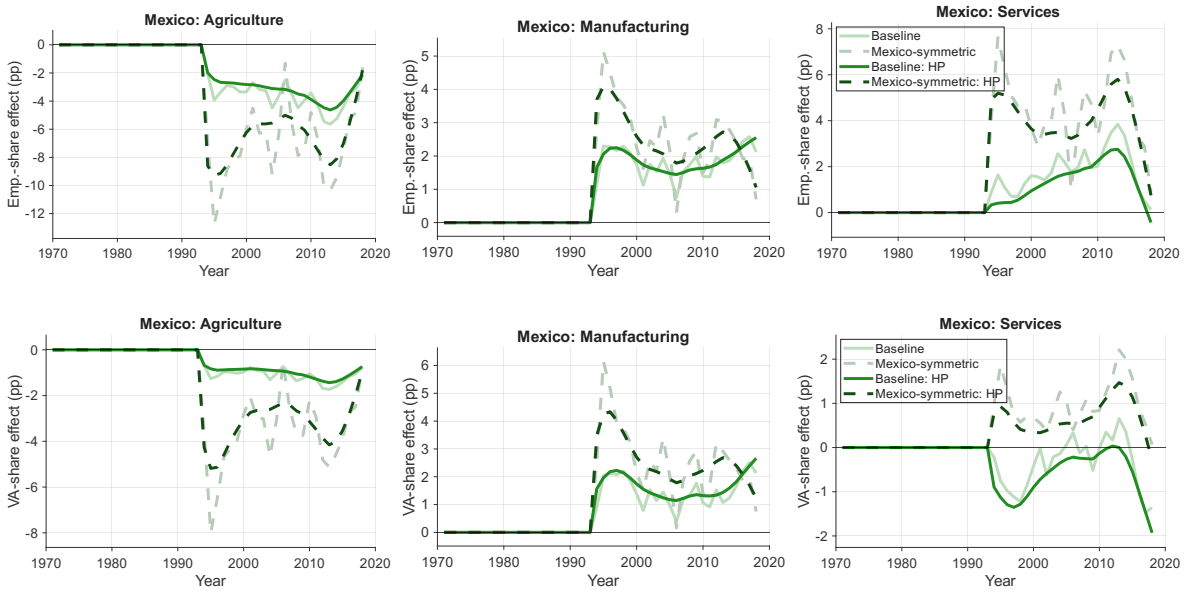


Figure A.7: NAFTA Effects by Development Environment

Notes: The figures plot NAFTA effects on employment and value-added shares in the baseline and *Mexico-symmetric*. The top row shows employment, while the bottom row shows value-added shares. Each effect is the active observed NAFTA-period trade-cost path minus the *no-NAFTA* frozen-path counterfactual. Solid lines report actual Mexico and dashed lines report the counterfactual development environment; lighter lines use annual calibrated primitives and darker lines use HP-filtered primitive simulations.

Table A.4: Mexico: Integration, GATT, and NAFTA Effects by Environment, 2010 and 2018

			Emp.-share effect (pp)			VA-share effect (pp)		
Shock	Environment	Year	Agr.	Mfg.	Svc.	Agr.	Mfg.	Svc.
Mexico integration	Baseline	2010	0.48	4.16	-4.64	0.06	5.76	-5.82
Mexico integration	Mexico-symmetric	2010	15.76	-1.59	-14.17	5.74	1.05	-6.79
Mexico integration	U.S./Canada pop. = Mexico	2010	0.71	3.34	-4.05	0.15	4.71	-4.86
GATT	Baseline	2010	-4.51	-1.28	5.79	-1.32	-2.90	4.22
GATT	Mexico-symmetric	2010	-0.15	-2.34	2.49	0.07	-3.88	3.81
GATT	U.S./Canada pop. = Mexico	2010	-3.09	-1.22	4.31	-0.89	-2.44	3.33
NAFTA	Baseline	2010	-3.41	1.39	2.02	-1.04	1.06	-0.02
NAFTA	Mexico-symmetric	2010	-4.82	1.49	3.32	-2.21	1.38	0.84
NAFTA	U.S./Canada pop. = Mexico	2010	-2.24	0.79	1.45	-0.68	0.54	0.14
Mexico integration	Baseline	2018	0.28	6.71	-6.99	0.02	8.49	-8.51
Mexico integration	Mexico-symmetric	2018	19.47	-1.35	-18.13	8.02	1.88	-9.90
Mexico integration	U.S./Canada pop. = Mexico	2018	0.38	5.45	-5.83	0.06	6.94	-7.00
GATT	Baseline	2018	-6.78	0.58	6.20	-2.34	-0.96	3.29
GATT	Mexico-symmetric	2018	3.06	-3.27	0.21	1.57	-4.23	2.66
GATT	U.S./Canada pop. = Mexico	2018	-5.70	0.22	5.48	-1.95	-1.10	3.06
NAFTA	Baseline	2018	-2.25	2.12	0.13	-0.77	2.13	-1.36
NAFTA	Mexico-symmetric	2018	-1.58	0.70	0.88	-0.81	0.75	0.07
NAFTA	U.S./Canada pop. = Mexico	2018	-1.62	1.30	0.31	-0.55	1.26	-0.71

Notes: Each effect is the active observed trade-cost path minus the corresponding *no-integration*, *no-GATT*, or *no-NAFTA* frozen-path counterfactual. The table reports Mexico under the baseline and *Mexico-symmetric*. The population-symmetric environment changes partner size only. Results use annual calibrated primitives.

B Solve Algorithm

B.1 Equilibrium conditions

Table A.5 provides a set of equations that characterize the equilibrium of the economy. This system of conditions jointly determines the endogenous allocation of labor, output, consumption, prices, and trade flows across countries and sectors, given exogenous fundamentals.

B.2 Algorithm

The model is solved as a sequence of static annual equilibria using the complete conditions in Table A.5. The corrected algorithm distinguishes destination expenditure, $X_{ik} = P_{ik}Q_{ik}$, from producer revenue, $R_{ik} = \sum_j \pi_{jik} X_{jk}$.

1. Take technologies, bilateral trade costs, labor endowments, transfer coefficients, and sectoral wage wedges as given. Initialize manufacturing wages and country incomes.

2. Given wages, solve the unit-cost and sectoral-price equations jointly using damping and a log-difference tolerance of 10^{-12}, and recover bilateral sourcing shares.

3. Compute household expenditure from candidate incomes and the transfer rule. Solve aggregate and sectoral consumption using the nonhomothetic demand system.

4. Stack country-sector expenditures, and let H contain intermediate-input requirements and S map destination expenditure into producer revenue. Solve the international system $X = (I - HS)^{-1}F$ and $R = SX$, where F is final expenditure. Intermediate demand therefore depends on domestic production, including exports.

5. Compute sectoral labor demand as $(1-\lambda_{ik})R_{ik}/(\xi_{ik}w_{im})$. Solve for wages and incomes using all national labor-market and income-consistency residuals together with the U.S. manufacturing-wage normalization. One income relation is redundant by world accounting; the Levenberg–Marquardt method retains it to control relative numerical accuracy in every country.

6. Require convergence, positive expenditures and allocations, and national labor and income consistency. No ex post labor rescaling is applied. Save the equilibrium and continue to the next year. The annual baseline and the complete counterfactual suite are recomputed using the same corrected solver and preference calibration.

Table A.5: Equilibrium Conditions

$D1:$	$C_{ik} = L_i \omega_{ik}^{\sigma} \left(\frac{P_{ik}}{P_i}\right)^{-\sigma} \left(\frac{C_i}{L_i}\right)^{(1-\sigma)\varepsilon_k+\sigma}$	$\forall i \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$D2:$	$\sum_{k\in\{a,m,s\}} P_{ik}C_{ik} = (1-\phi_i) \sum_{k\in\{a,m,s\}} w_{ik}L_{ik} + L_i T$	$\forall i \in \mathbb{I}$
$D3$	$P_i^{1-\sigma} = \sum_{k\in\{a,m,s\}} \omega_{ik}^{\sigma} P_{ik}^{1-\sigma} \left(\frac{C_i}{L_i}\right)^{(1-\sigma)(\varepsilon_k-1)}$	$\forall i \in \mathbb{I}$
$S1:$	$P_{ik} = G \left[\sum_{j=1}^{4} T_{jk} \left(v_{jk}\tau_{ijk}\right)^{-\theta}\right]^{-\frac{1}{\theta}}$	$\forall i \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$S2:$	$v_{ik} = \Upsilon_{ik} w_{ik}^{1-\lambda_{ik}} \left[\prod_{n=a,m,s} P_{in}^{\gamma_{ikn}}\right]^{\lambda_{ik}}$	$\forall i \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$S3:$	$\pi_{ijk} = \frac{T_{jk}\left(v_{jk}\tau_{ijk}\right)^{-\theta}}{\sum_{l=1}^{4} T_{lk}\left(v_{lk}\tau_{ilk}\right)^{-\theta}}$	$\forall i,j \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$S4:$	$L_{ik}w_{ik} = (1-\lambda_{ik}) \sum_{j=1}^{4} P_{jk}Q_{jk}\pi_{jik}$	$\forall i \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$S5:$	$w_{ik} = \xi_{ik} w_{im}$	$\forall i \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$M1:$	$L_i = \sum_{k\in\{a,m,s\}} L_{ik}$	$\forall i \in \mathbb{I}$
$M2:$	$Q_{ik} = C_{ik} + \sum_{n\in\{a,m,s\}} \gamma_{ink}\lambda_{in}\frac{R_{in}}{P_{ik}}$	$\forall i \in \mathbb{I},\ \forall k \in \{a,m,s\}$
$M3$	$\sum_{i=1}^{4} L_i \mathcal{T} = \sum_{i=1}^{4} \sum_{k\in\{a,m,s\}} \phi_i w_{ik} L_{ik}$	

Notes: $\mathbb{I} = \{1, 2, 3, 4\}$. Rows D, S, and M collect the demand, supply, and market-clearing conditions that define the model equilibrium. Given exogenous fundamentals, the system jointly determines sectoral consumption, prices, trade shares, output, labor allocations, and aggregate transfers. The numerical solution imposes the complete system: $D1$ and $D3$ enter the demand block, $D2$ and $M3$ determine household expenditure and the aggregate transfer, $M2$ maps final demand into gross output through the input–output system, and $S1$–$S5$ and $M1$ determine prices, trade shares, revenues, wages, and labor allocations.

C Data Appendix

This appendix documents the construction of the empirical database used in the quantitative analysis. It describes the primary data sources, the harmonization of countries and sectors, the construction of nominal and real aggregates, the aggregation of the rest-of-the-world economy, the treatment of bilateral trade flows and input–output coefficients, and the variables used to discipline preferences, productivity, trade costs, wage wedges, and counterfactual primitives.

C.1 Data Sources and Harmonization

The empirical database combines sectoral national accounts, employment, prices, bilateral trade flows, input-output linkages, population, income, and purchasing-power-parity information for the period 1971–2018. The construction proceeds in two conceptual steps. First, each raw source is harmonized to common country identifiers, annual frequency, and the three-sector classification used in the model. Second, the harmonized sources are combined into a single country-sector panel using source-priority rules that reflect coverage, internal consistency, and proximity to the original statistical agency. Nominal variables are expressed in current U.S. dollars; real variables are benchmarked to 2005 PPP prices.

The main sources are EU KLEMS, the World Input-Output Database, the OECD Inter-Country Input-Output tables, the Groningen Growth and Development Centre (GGDC) 10-Sector Database, the Groningen Productivity Level Database, the U.S. Bureau of Economic Analysis, Statistics Canada, United Nations sectoral value-added data, the World Development Indicators, and the Penn World Table. These sources are used jointly because no single database provides consistent long-run coverage of employment, value added, gross output, intermediate inputs, bilateral trade, and PPP-adjusted real quantities for all countries in the sample.

C.2 Country Sample and Sector Mapping

The model contains four economies: Mexico, Canada, the United States, and a rest-of-the-world aggregate. The rest of the world consists of Australia, Austria, Belgium, Brazil, China, Germany, Denmark, Spain, Finland, France, the United Kingdom, Greece, India, Ireland, Italy, Japan, South Korea, Luxembourg, the Netherlands, Portugal, Sweden, and Taiwan. This country set is chosen to maximize the overlap between long-run sectoral accounts, input-output information, and bilateral trade data.

All observations are mapped into three broad sectors. Agriculture includes agriculture, forestry, and fishing. Manufacturing includes mining, manufacturing, utilities, and construction. Services include all remaining market and non-market activities. The mapping is implemented at the most disaggregated level available in each source, using the relevant ISIC, NACE, WIOD, ICIO, and GGDC classifications before aggregation to the three model sectors.

C.3 National Accounts, Prices, and Real Aggregates

Sectoral employment, nominal value added, real value added, nominal gross output, real gross output, and intermediate inputs are assembled from the harmonized national-account sources. When multiple sources overlap, the preferred source is selected using the following hierarchy. Country-specific statistical agencies are used to fill employment gaps for the United States and Canada. EU KLEMS and WIOD provide the core internationally comparable sectoral accounts for advanced economies. GGDC extends coverage for economies and periods with incomplete EU KLEMS or WIOD coverage. United Nations value-added series fill selected remaining gaps, especially where long historical coverage is otherwise unavailable. Intermediate inputs are measured directly when possible and are otherwise inferred as gross output less value added.

Real sectoral quantities are aggregated using Tornqvist growth rates. Within each country and broad sector, industry expenditure shares are used to weight industry-level real growth rates.

The sectoral real level is then recovered by applying the resulting growth rates forward and backward from the relevant source benchmark year. This procedure avoids simple summation of real quantities outside the benchmark year and preserves changes in the relative importance of detailed industries within each broad sector.

Nominal series are converted into U.S. dollars using exchange rates derived from national-currency and U.S.-dollar GDP series in the World Development Indicators, with a separate exchange-rate series for Taiwan. Real value added, real gross output, and real intermediate inputs are benchmarked to 2005 PPP prices using conversion factors constructed from the Groningen Productivity Level Database. The gross-output price index used in the calibration is

$$P^{GO}_{ik,t} = \frac{GO^{USD}_{ik,t}}{GO^{PPP05}_{ik,t}},$$

and an analogous value-added price index is used as a diagnostic series. Missing observations are filled with source-specific growth rates when reliable adjacent observations are available. When no adjacent observation exists, the series is backcast or extended using average annual growth over the nearest reliable window.

C.4 Rest-of-the-World Aggregate

The rest-of-the-world aggregate is formed after country-level series have been expressed in common units. Employment and nominal variables are summed across the countries assigned to the aggregate. Real quantities are constructed from country-sector growth rates using Tornqvist aggregation and 2005 PPP benchmarks. Price indexes for the rest of the world are output-weighted averages within each year and sector, with weights based on U.S.-dollar value added or gross output, depending on the object being aggregated.

C.5 Bilateral Trade and Input-Output Coefficients

Bilateral trade flows are constructed from WIOD and OECD ICIO trade data. Origin and destination industries are mapped into the three model sectors, and all trade involving economies outside Mexico, Canada, and the United States is assigned to the rest-of-the-world aggregate. The resulting object is a set of annual destination-origin trade matrices by sector. For goods sectors, the historical and post-1995 trade series are combined to obtain long-run coverage. For services, bilateral flows are constructed from services imports and exports in the World Development Indicators, with missing values interpolated or backcast using country-specific growth rates. Foreign services imports are allocated across origins using observed services-trade shares, and domestic services absorption is measured as gross output plus imports minus exports.

Because comprehensive bilateral, sector-level services-trade data are unavailable for large stretches of the 1971–2018 sample, a non-trivial share of the services-trade matrix relies on the allocation and interpolation procedures described above rather than on directly reported bilateral flows. This measurement limitation should be kept in mind when interpreting the services-sector counterfactual results in Section 5; I treat the tradable-services specification adopted here as the paper's benchmark modeling choice rather than as an uncontested one.

C.6 Population, Income Gaps, and Calibration Variables

Population and real GDP per capita are taken from the Penn World Table (Feenstra et al. 2016). The income-gap target is the 1971 population-weighted real GDP per capita of each model economy relative to Mexico. It is used as a proxy for relative model consumption per employed worker; the population and employed-labor denominators are distinct.

C.7 Aggregation Procedure

Employment and all nominal variables are aggregated to the sector level by summation. For real variables, simple summation is exact only in the reference year, when nominal and real values coincide. For all other years, sectoral real quantities are constructed via Tornqvist indexes applied to the reference-year benchmarks. Specifically, for any variable x, let $x_n^j(t)$ denote the real quantity in industry $j \in \{1, \ldots, N_n\}$ within sector n at time t, and $X_n(t)$ the sectoral aggregate real value. The growth rate of $X_n(t)$ is:

$$\Delta \log X_n(t) = \sum_{j=1}^{N_n} \bar{s}_n^j(t) \left(\log x_n^j(t) - \log x_n^j(t-1) \right), \tag{A.1}$$

where

$$s_n^j(t) \equiv \frac{p_{x,n}^j(t)\, x_n^j(t)}{\sum_{i=1}^{N_n} p_{x,n}^i(t)\, x_n^i(t)}, \qquad \bar{s}_n^j(t) \equiv \tfrac{1}{2}\left(s_n^j(t) + s_n^j(t-1) \right),$$

with $p_{x,n}^j(t)$ the price index for industry j in sector n. Sectoral level series are then recovered by applying these growth rates to the reference-year benchmark values.